RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

Rajalakshmi Nagar, Thandalam – 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS

LABORATORY RECORD

Submitted by :	DEEPAN S A
Register Number :	241901021
Branch :	B.E. Computer Science and Engineering (Cyber Security)
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Faculty In-charge :	Dr. D. SIVAKUMAR
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EXPERIMENT 1

Study of Computer Forensics and Different Tools Used for Forensic Investigation

Aim

To study the fundamentals of computer forensics and explore various commercial, open-source, and field tools used in digital crime investigations.

Overview of Digital Forensics

- **Concept:** Digital forensics is the field of identifying, collecting, examining, and analyzing digital evidence to determine responsibility for cyber intrusions or computer crimes.
- **Core Principle:** Perpetrators leave digital traces (log entries, registry modifications, deleted files, malware remnants) that serve as investigative evidence.



Result

The study of computer forensics concepts and various forensic investigation tools was successfully completed.

EXPERIMENT 02

Live Forensics Case Investigation using Autopsy

Aim

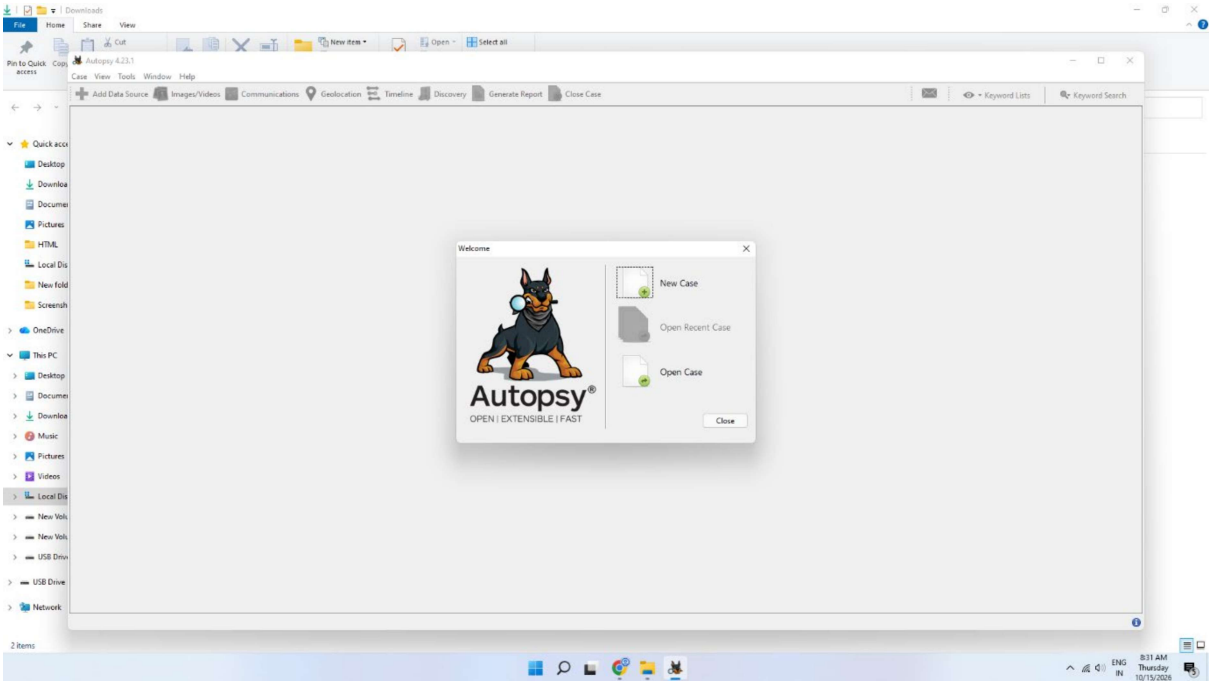
To perform a live digital forensics case investigation and analyze system evidence using the Autopsy forensic tool.

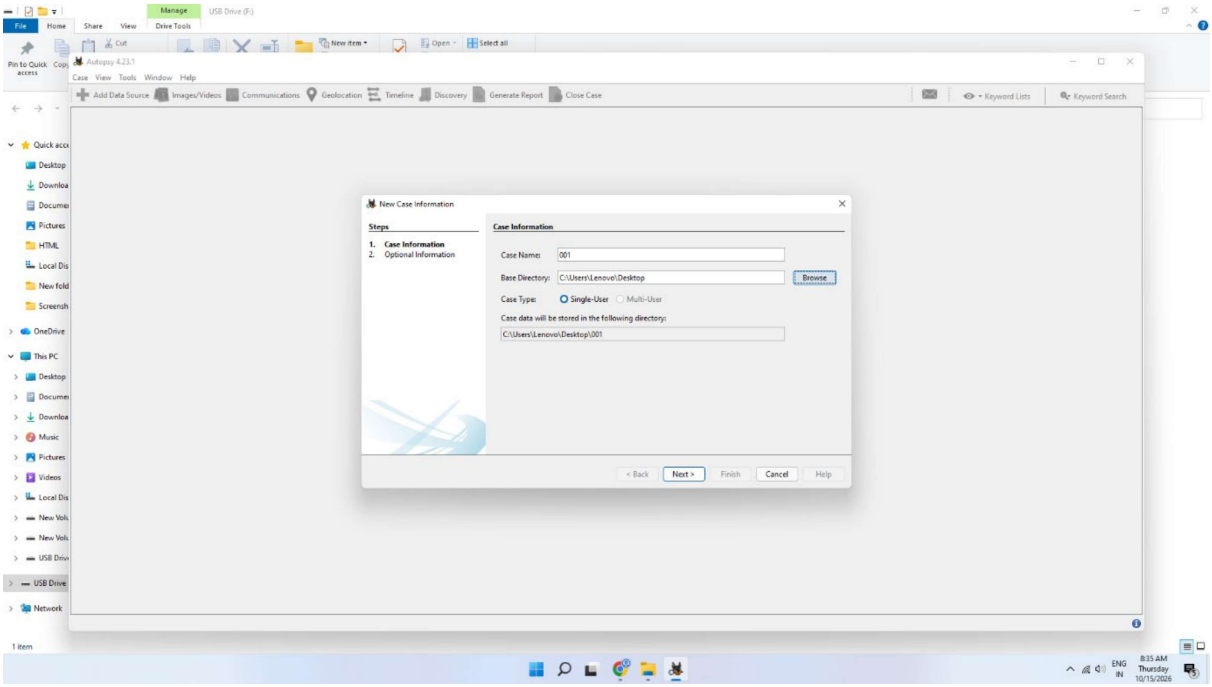
Software / Tools Required

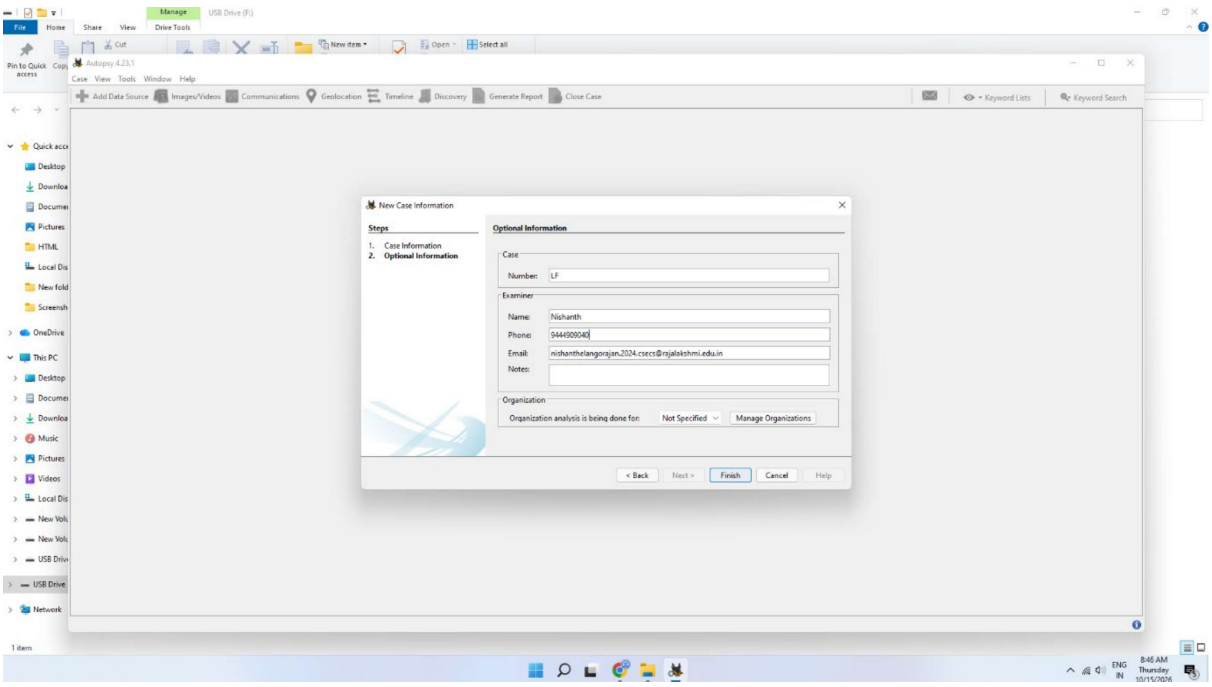
- Autopsy Digital Forensics Platform
- Target System / Evidence Image / Data Source

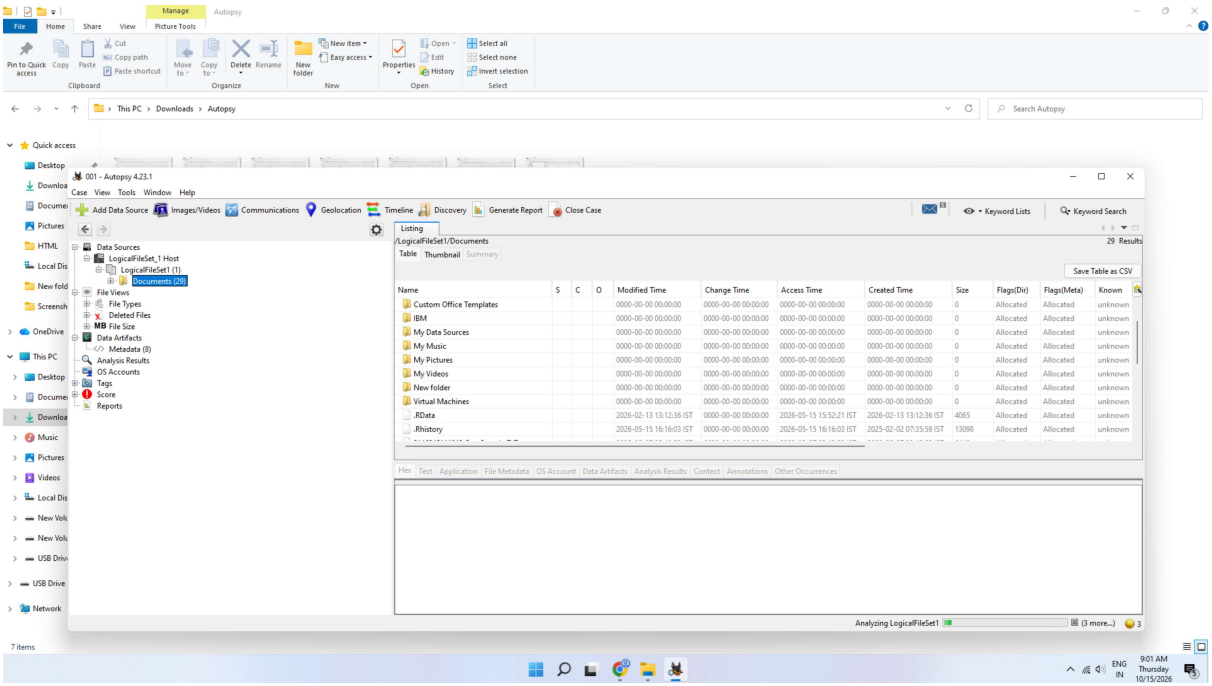
Procedure

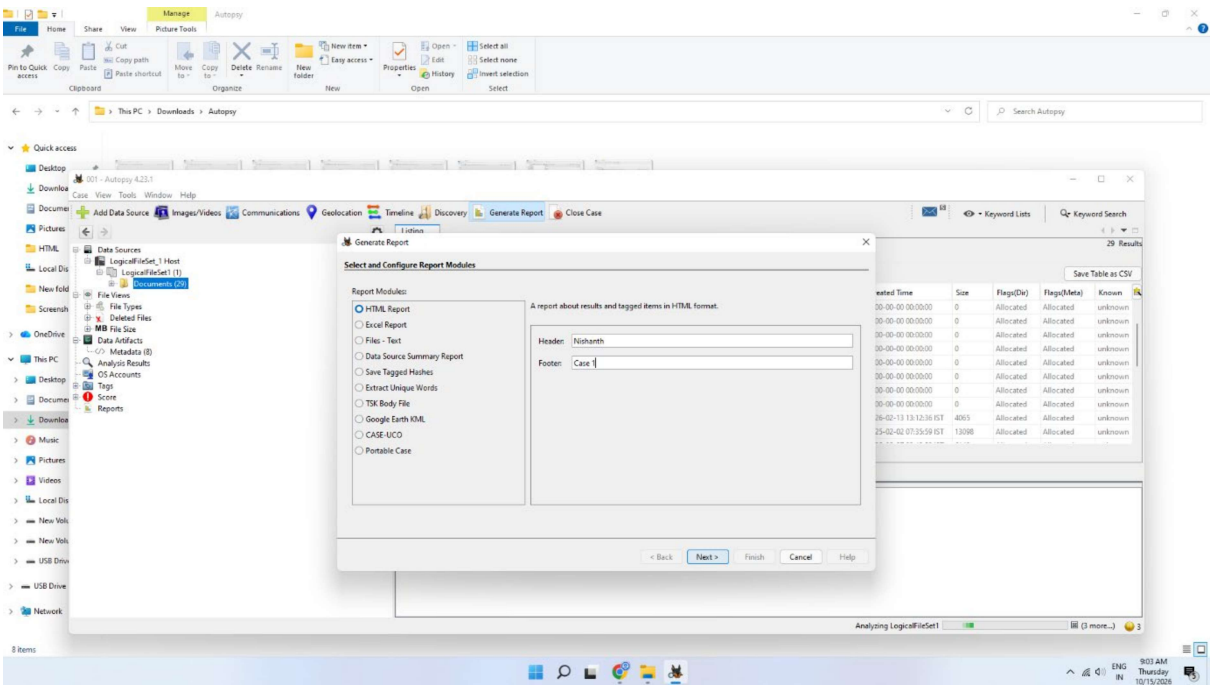
1. **Case Creation:** Launched Autopsy, selected **New Case**, and configured the *Case Name*, *Base Directory*, *Case Number*, and *Examiner Details*.
2. **Data Source Addition:** Added the target drive/disk path as the data source to the local database.
3. **Ingest Modules Configuration:** Enabled necessary ingest modules to scan and extract evidence.
4. **Evidence Analysis:** Processed and navigated through the generated investigation result.

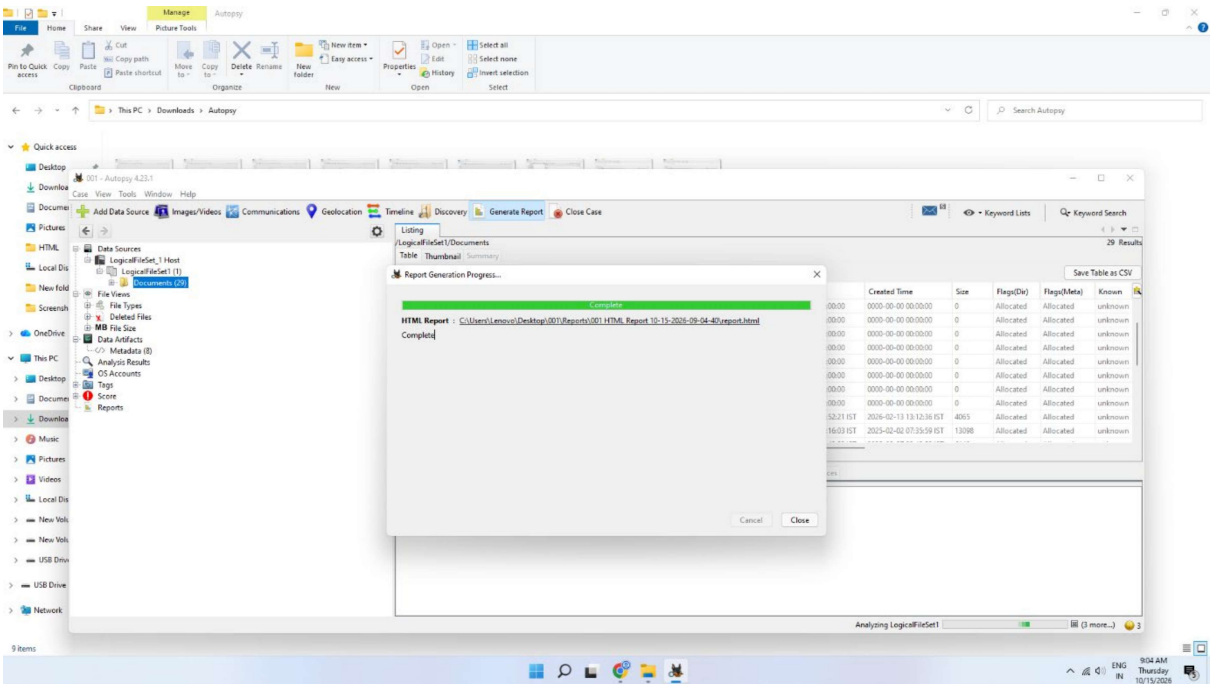


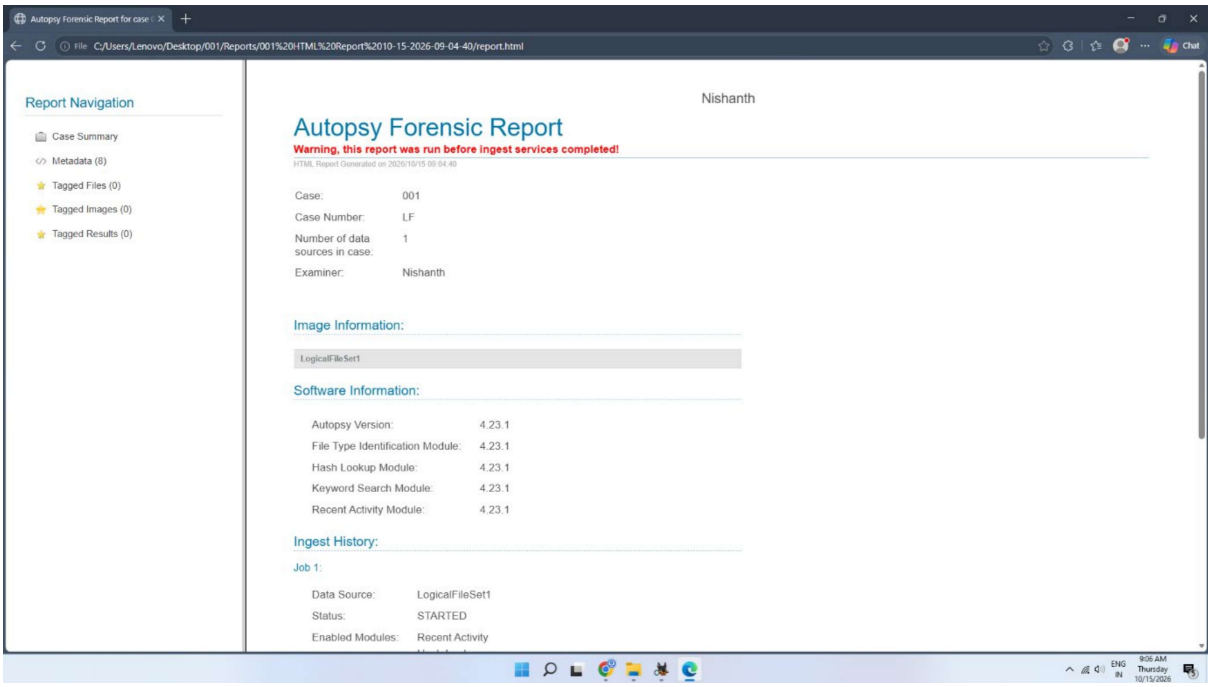

New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: 001
Base Directory: C:\Users\Lenovo\Desktop
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
C:\Users\Lenovo\Desktop\001
< Back
Next >
Finish
Cancel
Help


New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number: LF
Examiner
Name: Nishanth
Phone: 9444909040
Email: nishanthelangorajan.2024.csecs@rajalakshmi.edu.in
Notes:
Organization
Organization analysis is being done for: Not Specified
Manage Organizations
< Back
Next >
Finish
Cancel
Help


001 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
LogicalFileSet_1 Host
LogicalFileSet1 (1)
Documents (29)
File Views
File Types
Deleted Files
File Size
Data Artifacts
Metadata (8)
Analysis Results
OS Accounts
Tags
Score
Reports
Listing
/LogicalFileSet1/Documents
29 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
Flags(Meta)
Known
Custom Office Templates
IBM
My Data Sources
My Music
My Pictures
My Videos
New folder
Virtual Machines
.RData
.Rhistory
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Analyzing LogicalFileSet1
7 items


001 - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Generate Report
Select and Configure Report Modules
Report Modules:
HTML Report
Excel Report
Files - Text
Data Source Summary Report
Save Tagged Hashes
Extract Unique Words
TSK Body File
Google Earth KML
CASE-UCO
Portable Case
A report about results and tagged items in HTML format.
Header:
Nishanth
Footer:
Case 1
< Back
Next >
Finish
Cancel
Help
Analyzing LogicalFileSet1
8 items





Result

The Autopsy tool was successfully executed, and digital evidence was captured, categorized, and analyzed.

EXPERIMENT 03

Recover Deleted Files using Forensics Tools

Aim

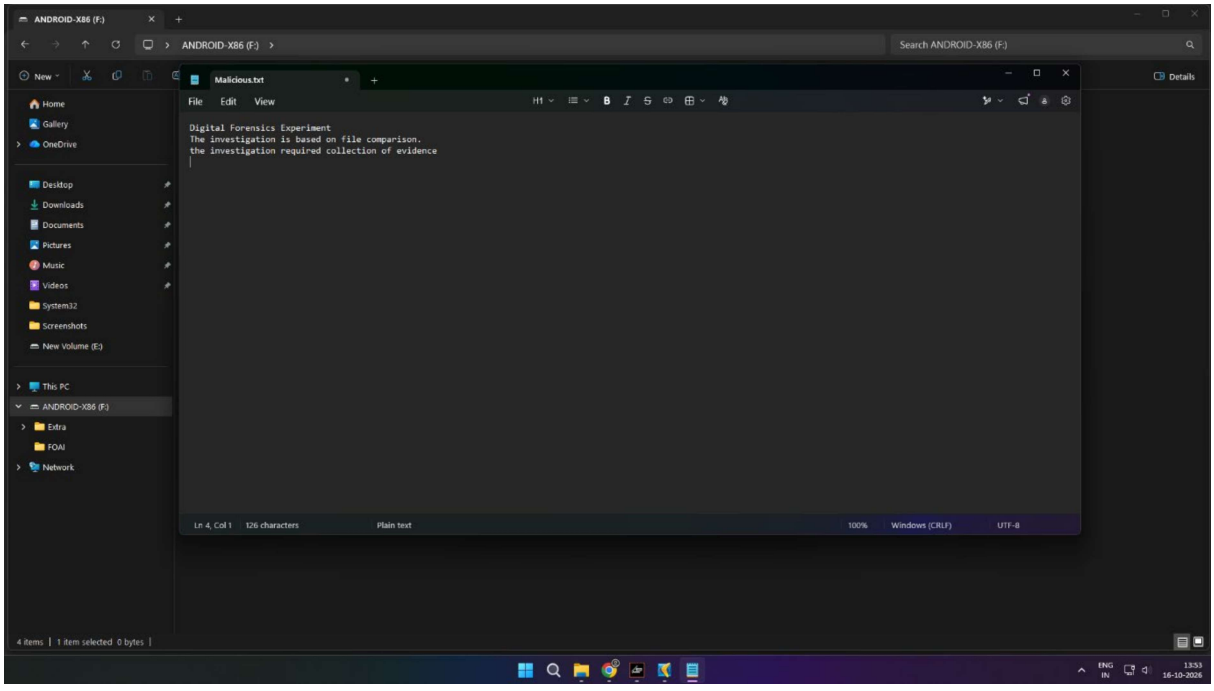
To practically demonstrate the process of recovering deleted files from a drive using digital forensic imaging and recovery tools.

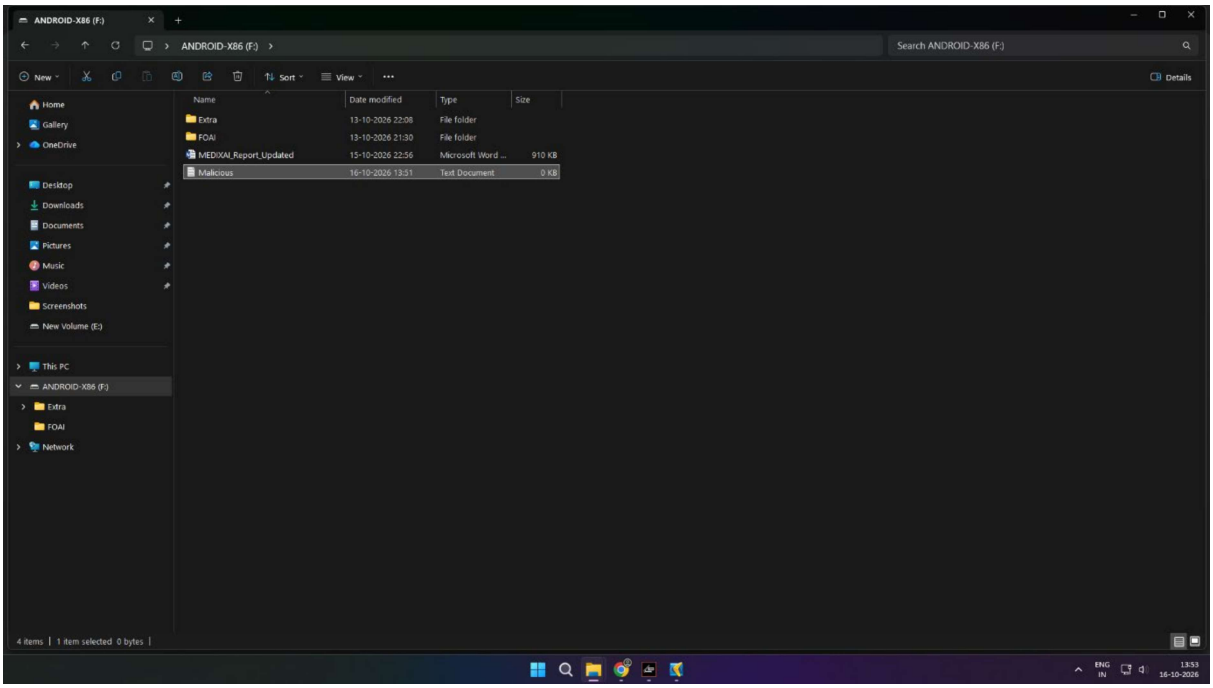
Software / Tools Required

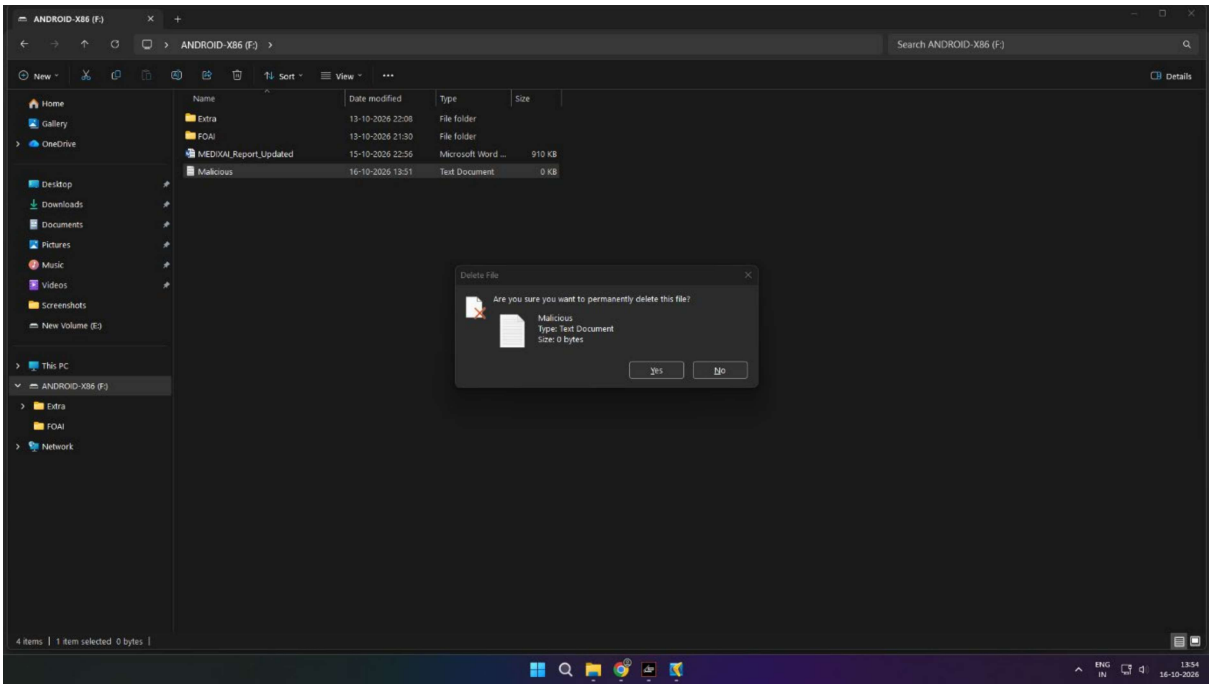
- **FTK Imager** (for bit-by-bit imaging)
- **RecoverMyFiles** (for scanning and recovery)
- Target Storage Device (e.g., USB Flash Drive)

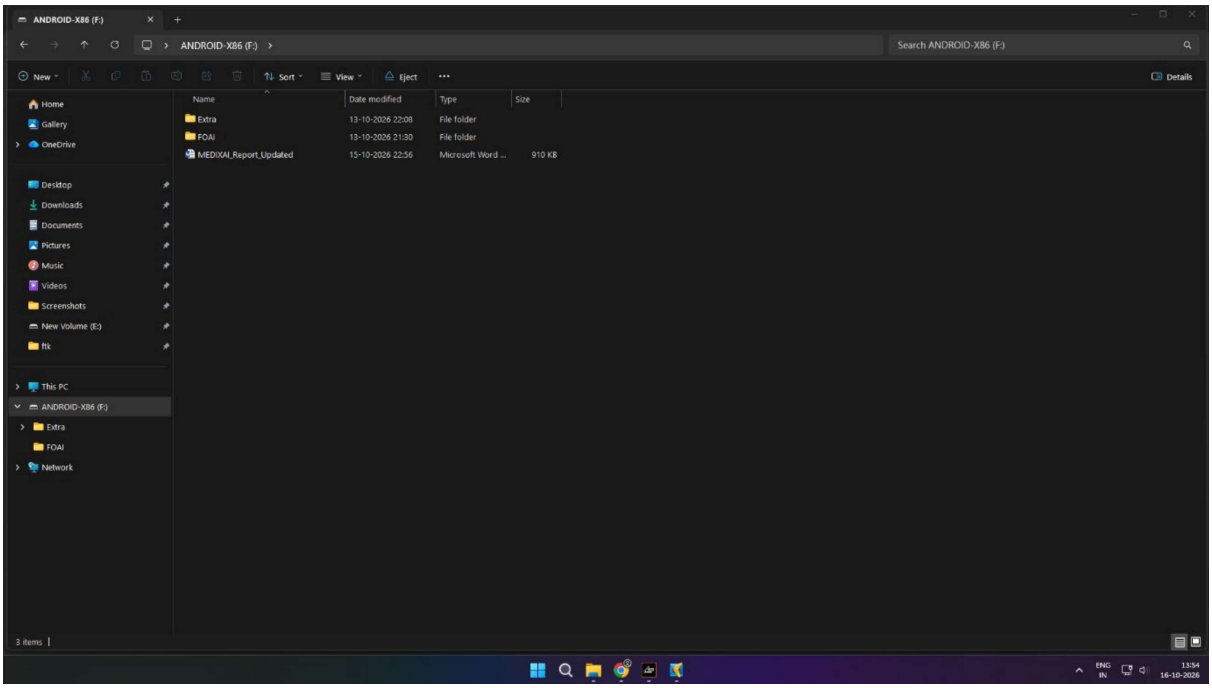
Procedure

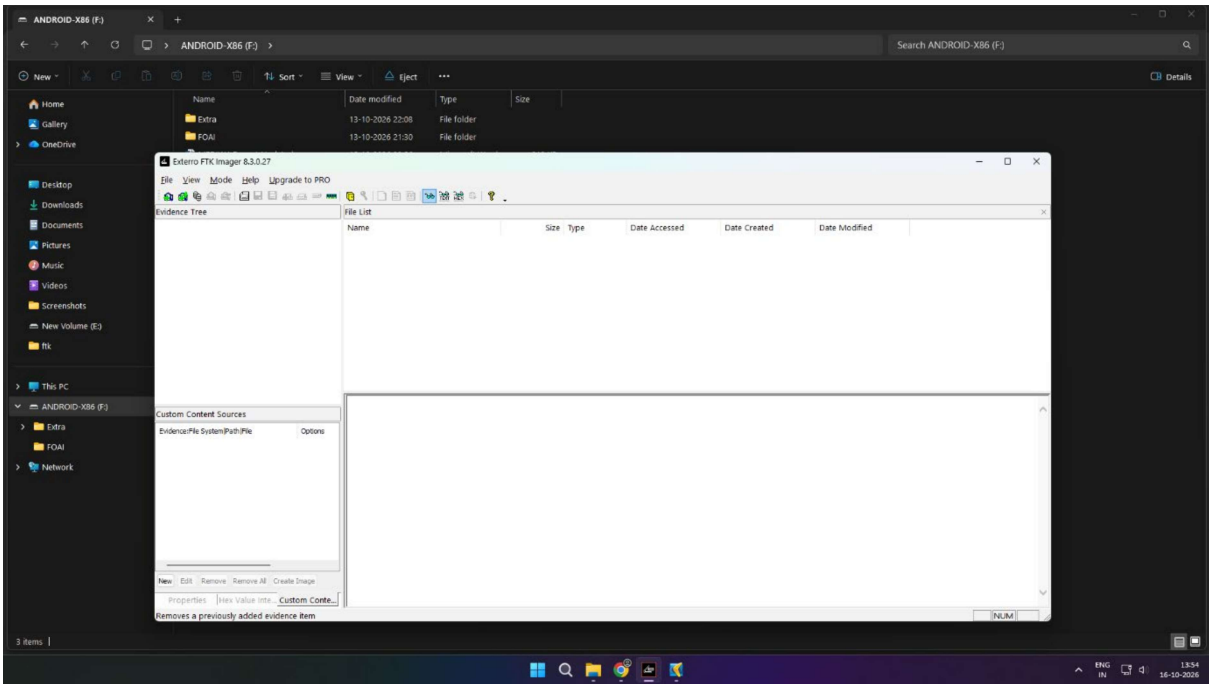
1. **Evidence Simulation:** Created a test text file (e.g., malicious.txt) on a flash drive and intentionally deleted it.
2. **Drive Imaging:** Used **FTK Imager** to select the physical drive and create a bit-by-bit forensic image.
3. **Data Recovery:** Opened **RecoverMyFiles**, selected "Recover a Drive", and loaded the created forensic image.
4. **File Extraction:** Executed the file recovery scan, filtered results by file type, and located the deleted file.

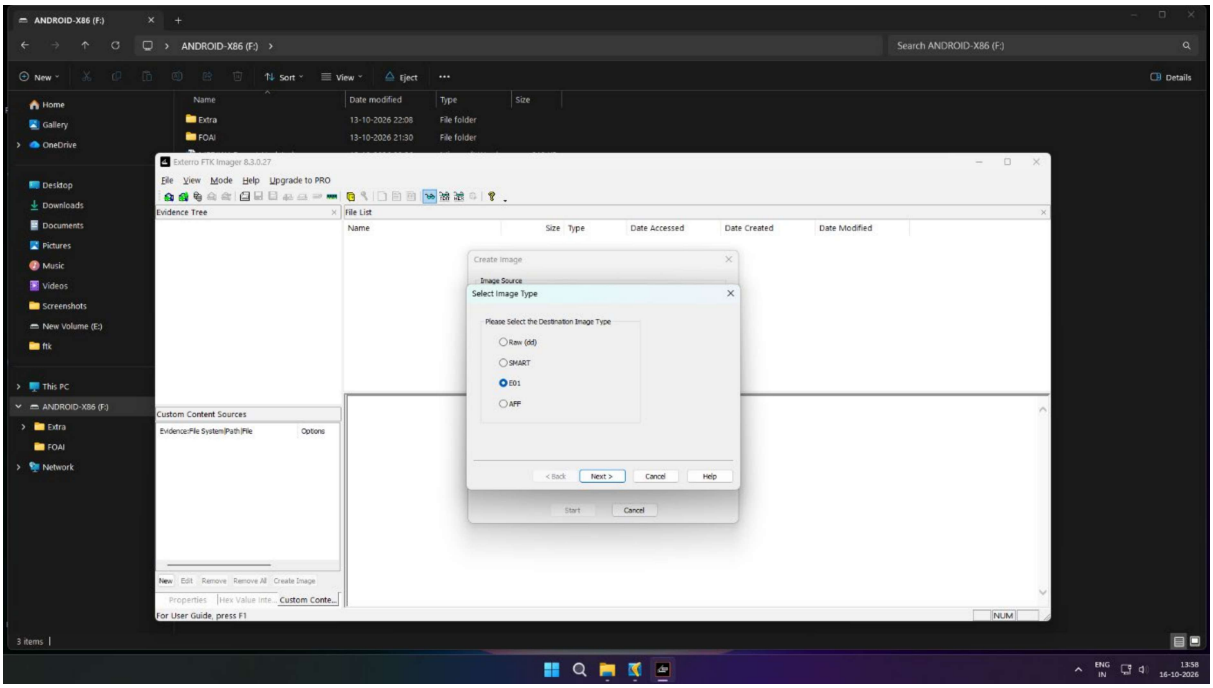


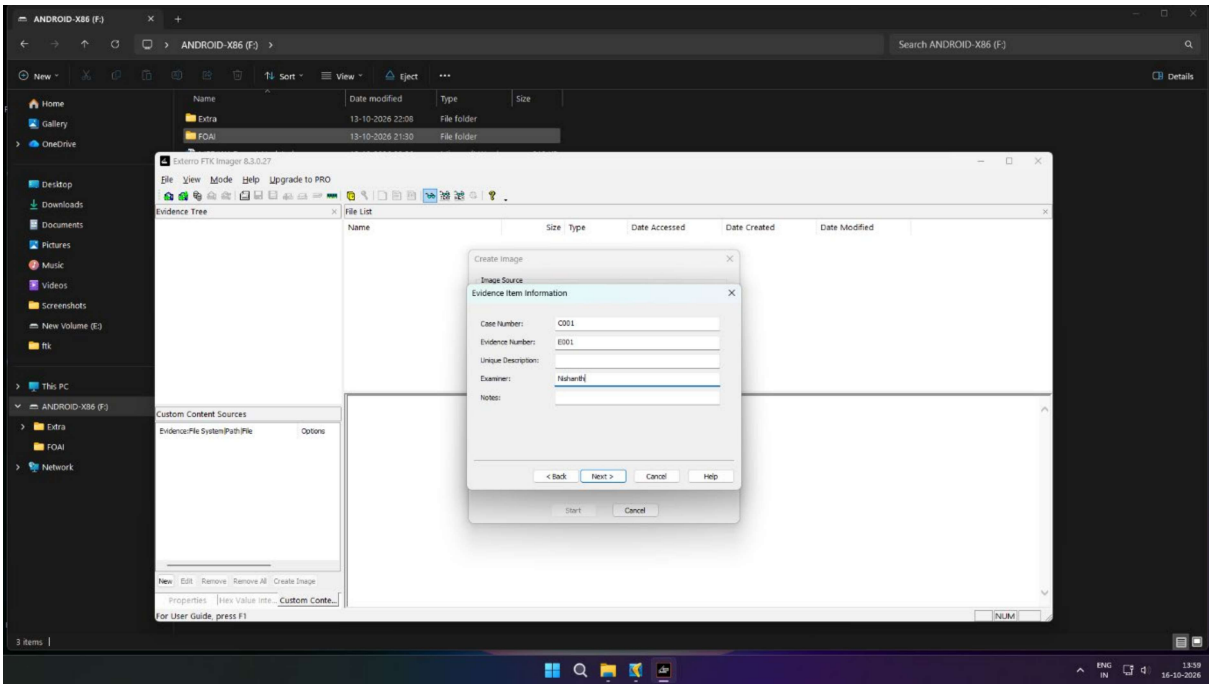

ANDROID-X86 (F:)
Search ANDROID-X86 (F:)
New
Sort
View
Details
Home
Gallery
OneDrive
Desktop
Downloads
Documents
Pictures
Music
Videos
Screenshots
New Volume (E:)
This PC
ANDROID-X86 (F:)
Extra
FOAI
Network
Name
Date modified
Type
Size
Extra 13-10-2026 22:08 File folder
FOAI 13-10-2026 21:30 File folder
MEDIXAI_Report_Updated 15-10-2026 22:56 Microsoft Word ... 910 KB
Malicious 16-10-2026 13:51 Text Document 0 KB
4 items | 1 item selected 0 bytes
ENG IN
13:53 16-10-2026

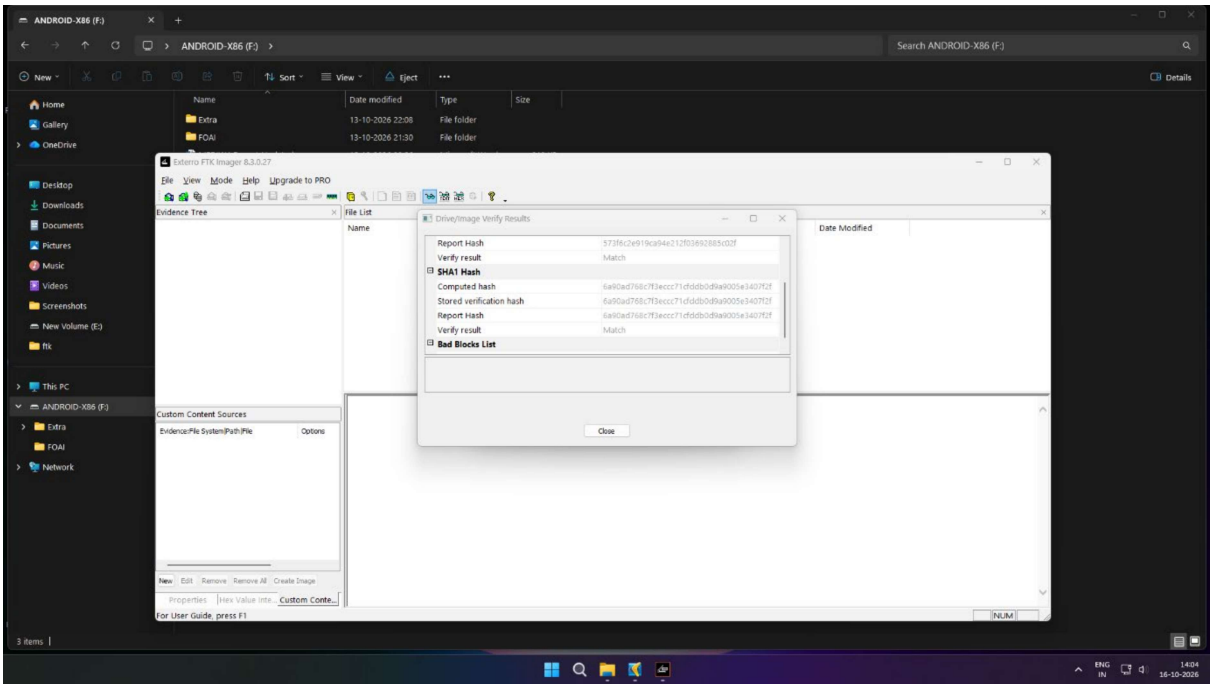

ANDROID-X86 (F:)
Search ANDROID-X86 (F:)
New
Sort
View
Details
Home
Gallery
OneDrive
Desktop
Downloads
Documents
Pictures
Music
Videos
Screenshots
New Volume (E:)
This PC
ANDROID-X86 (F:)
Extra
FOAI
Network
Name
Date modified
Type
Size
Extra 13-10-2026 22:08 File folder
FOAI 13-10-2026 21:30 File folder
MEDIXAI_Report_Updated 15-10-2026 22:56 Microsoft Word ... 910 KB
Malicious 16-10-2026 13:51 Text Document 0 KB
Delete File
Are you sure you want to permanently delete this file?
Malicious
Type: Text Document
Size: 0 bytes
Yes
No
4 items | 1 item selected 0 bytes
ENG IN
13:54 16-10-2026

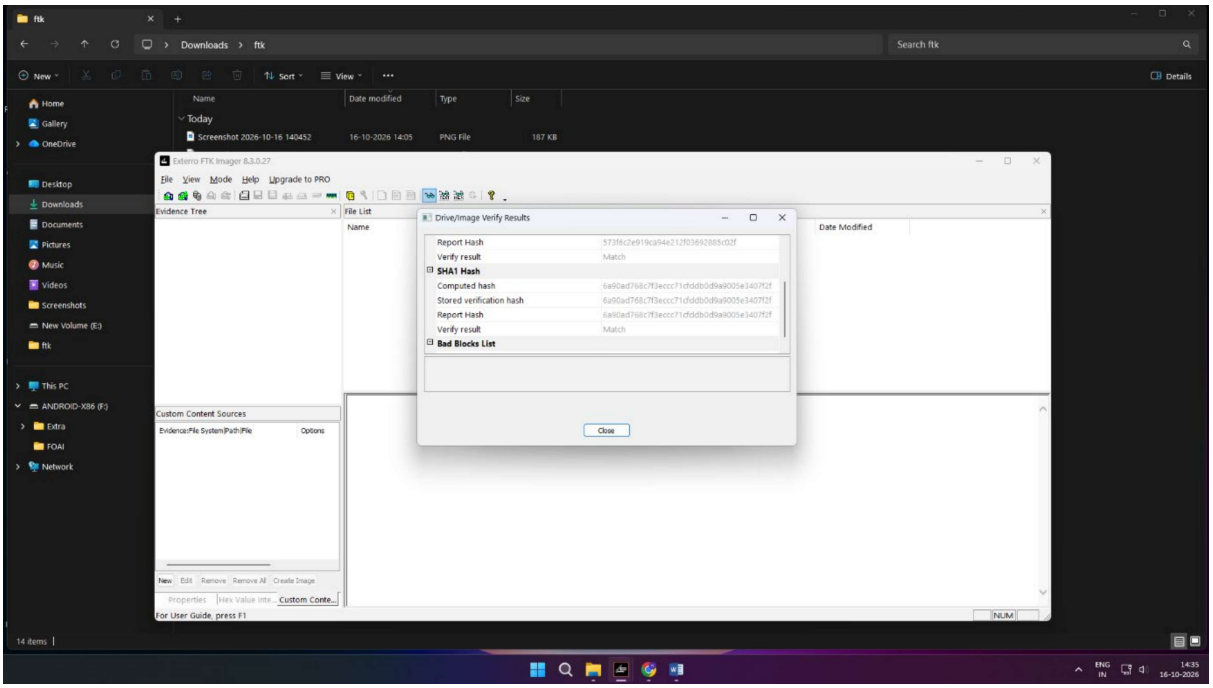

ANDROID-X86 (F:)
Search ANDROID-X86 (F:)
New
Sort
View
Eject
Details
Home
Gallery
OneDrive
Desktop
Downloads
Documents
Pictures
Music
Videos
Screenshots
New Volume (E:)
ftk
This PC
ANDROID-X86 (F:)
Extra
FOAI
Network
Name
Date modified
Type
Size
Extra
13-10-2026 22:08
File folder
FOAI
13-10-2026 21:30
File folder
MEDIXAI_Report_Updated
13-10-2026 22:56
Microsoft Word ...
910 KB
3 items
ENG
IN
13:54
16-10-2026

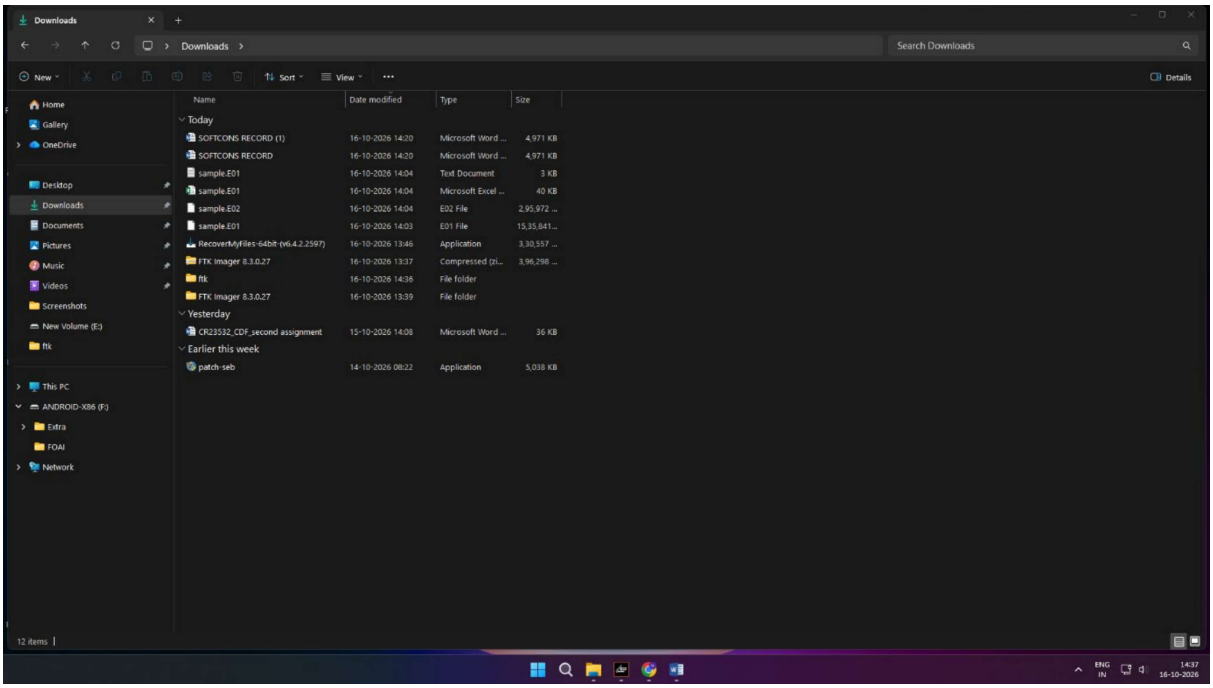

ANDROID-X86 (F:)
Search ANDROID-X86 (F:)
New
Sort
View
Eject
Details
Home
Gallery
OneDrive
Desktop
Downloads
Documents
Pictures
Music
Videos
Screenshots
New Volume (E:)
ftk
This PC
ANDROID-X86 (F:)
Extra
FOAI
Network
Name
Date modified
Type
Size
Extra
13-10-2026 22:08
File folder
FOAI
13-10-2026 21:30
File folder
Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
File List
Name
Size
Type
Date Accessed
Date Created
Date Modified
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value Inte...
Custom Conte...
Removes a previously added evidence item
NUM
3 items
ENG
IN
13:54
16-10-2026

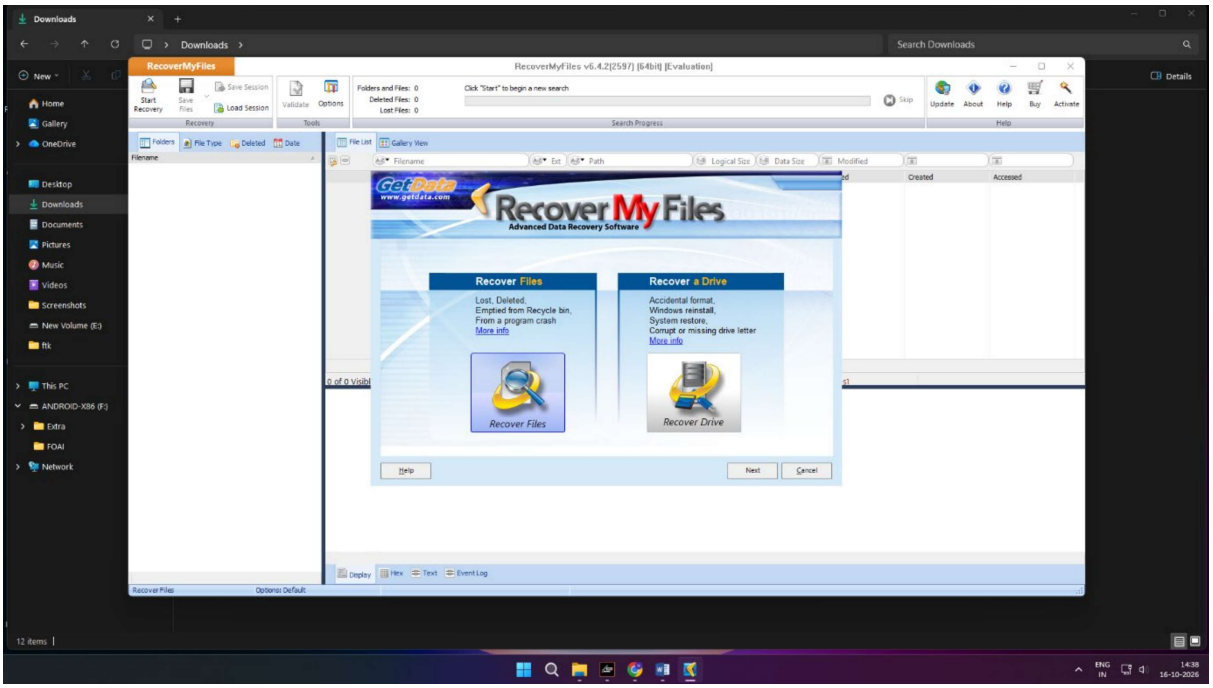

ANDROID-X86 (F:)
Search ANDROID-X86 (F:)
Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
File List
Create Image
Image Source
Select Image Type
Please Select the Destination Image Type
Raw (dd)
SMART
E01
AFF
< Back
Next >
Cancel
Help
Start
Custom Content Sources
For User Guide, press F1

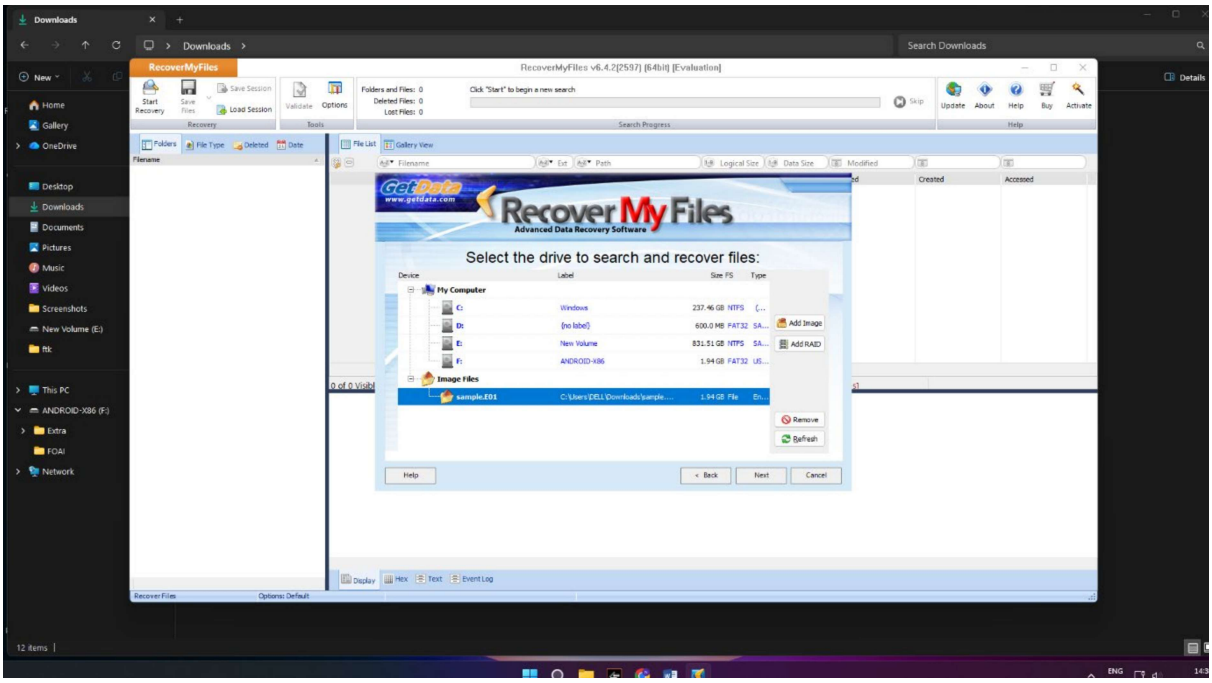

ANDROID-X86 (F:)
Search ANDROID-X86 (F:)
Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
File List
Create Image
Image Source
Evidence Item Information
Case Number:
C001
Evidence Number:
E001
Unique Description:
Examiner:
Nishanth
Notes:
< Back
Next >
Cancel
Help
Start
Custom Content Sources
For User Guide, press F1

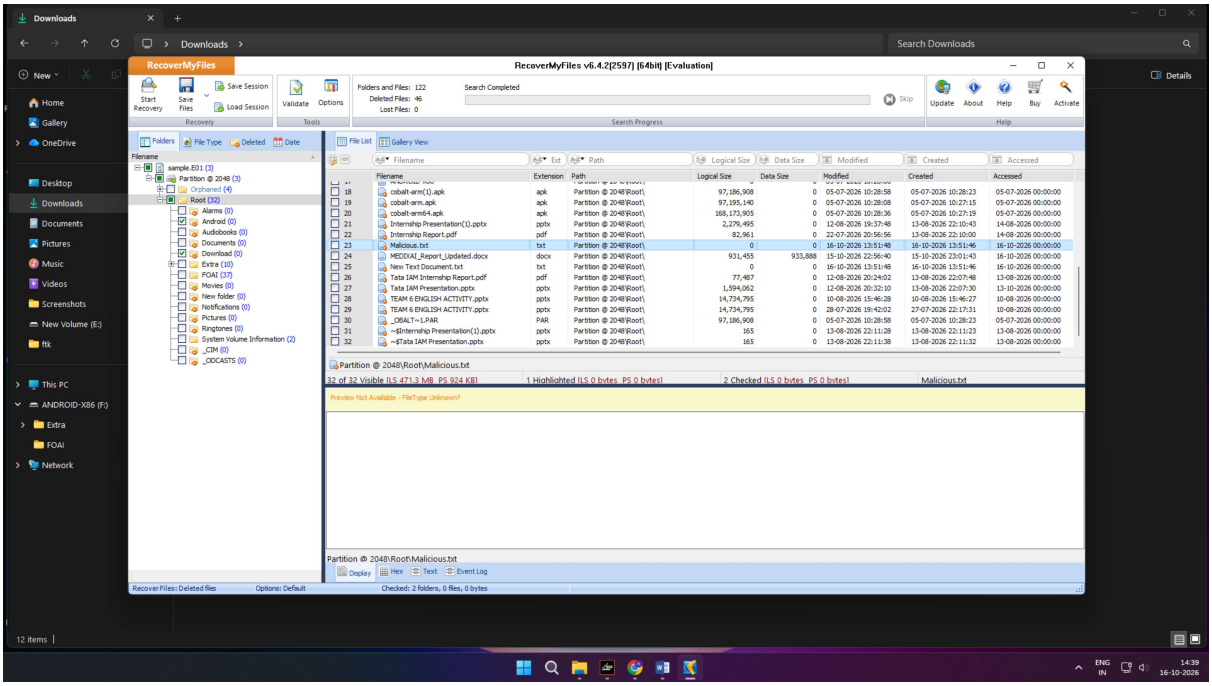

ANDROID-X86 (F:)
Search ANDROID-X86 (F:)
Exterro FTK Imager 8.3.0.27
Drive/Image Verify Results
Report Hash
Verify result
SHA1 Hash
Computed hash
Stored verification hash
Bad Blocks List
Close


ftk
Downloads › ftk
Search ftk
Exterro FTK Imager 8.3.0.27
Drive/Image Verify Results
Report Hash
Verify result
SHA1 Hash
Computed hash
Stored verification hash
Bad Blocks List
Close


Downloads
Search Downloads
New
Sort
View
Details
Home
Gallery
OneDrive
Desktop
Downloads
Documents
Pictures
Music
Videos
Screenshots
New Volume (E:)
ftk
This PC
ANDROID-X86 (F:)
Extra
FOAI
Network
Name
Date modified
Type
Size
Today
SOFTCONS RECORD (1) 16-10-2026 14:20 Microsoft Word ... 4,971 KB
SOFTCONS RECORD 16-10-2026 14:20 Microsoft Word ... 4,971 KB
sample.E01 16-10-2026 14:04 Text Document 3 KB
sample.E01 16-10-2026 14:04 Microsoft Excel ... 40 KB
sample.E02 16-10-2026 14:04 E02 File 2,95,972 ...
sample.E01 16-10-2026 14:03 E01 File 15,35,841...
RecoverMyFiles-64bit-(v6.4.2.2597) 16-10-2026 13:46 Application 3,30,557 ...
FTK Imager 8.3.0.27 16-10-2026 13:37 Compressed (zi... 3,96,298 ...
ftk 16-10-2026 14:36 File folder
FTK Imager 8.3.0.27 16-10-2026 13:39 File folder
Yesterday
CR23532_CDF_second assignment 15-10-2026 14:08 Microsoft Word ... 36 KB
Earlier this week
patch-seb 14-10-2026 08:22 Application 5,038 KB
12 items
ENG
IN
14:37
16-10-2026


Downloads
Search Downloads
RecoverMyFiles
RecoverMyFiles v6.4.2(2597) [64bit] [Evaluation]
Start Recovery
Save Files
Save Session
Load Session
Validate
Options
Folders and Files: 0
Deleted Files: 0
Lost Files: 0
Click "Start" to begin a new search
Skip
Update
About
Help
Buy
Activate
Recovery
Tools
Search Progress
Help
Folders
File Type
Deleted
Data
File List
Gallery View
Filename
Ext
Path
Logical Size
Data Size
Modified
Created
Accessed
GetData
www.getdata.com
Recover My Files
Advanced Data Recovery Software
Recover Files
Lost, Deleted,
Emptied from Recycle bin,
From a program crash
More info
Recover Files
Recover a Drive
Accidental format,
Windows reinstall,
System restore,
Corrupt or missing drive letter
More info
Recover Drive
Help
Next
Cancel
Display
Hex
Text
Event Log
Recover Files
Options: Default
12 items
ENG
IN
14:38
16-10-2026





Result

The intentionally deleted digital evidence was accurately captured, analyzed, and recovered.

EXPERIMENT 04

Find Last Connected USB on Your System (USB Forensics)

Aim

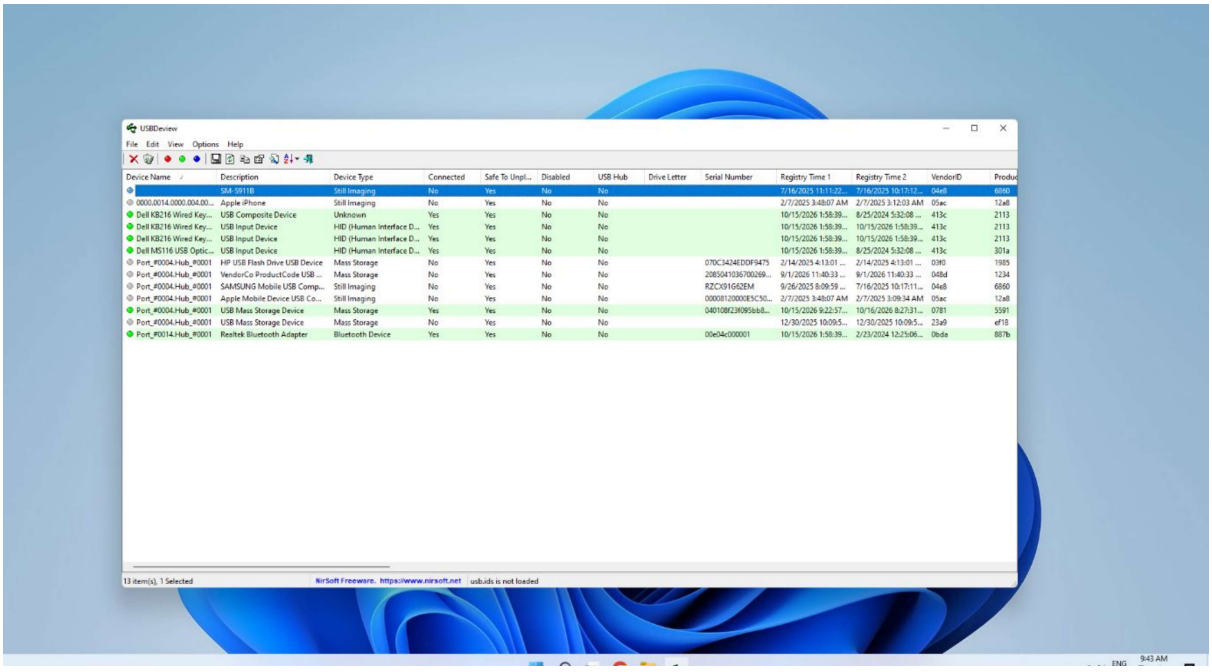
To identify and analyze history, connection logs, and details of USB devices previously or currently connected to the system.

Software / Tools Required

- USBDeview Utility
- Target Windows Host PC

Procedure

1. Executed **USBDeview** on the target Windows system with administrator privileges.
2. Retrived the generated list of all currently connected and previously attached USB devices.
3. Extracted device attributes (Device Name, Serial Number, Vendor ID, Product ID, and Last Plug/Unplug Timestamps).



USB Devices List

Created by using USBDeview

Device Name	Description	Device Type	Connected	Safe To Unplug	Disabled	USB Hub	Drive Letter	Serial Number
0000.0014.0000.001.000.000.000.000.000	MTP	Still Imaging	No	No	No	No		
0000.0014.0000.001.000.000.000.000.000	USB Serial Device	Communication	No	No	No	No	COM3	
0000.0014.0000.004.000.000.000.000.000	Apple iPhone	Still Imaging	No	Yes	No	No		
0000.0014.0000.004.000.000.000.000.000	Apple iPhone	Still Imaging	No	Yes	No	No		
0000.0014.0000.007.000.000.000.000.000	USB Input Device	HID (Human Interface Device)	No	Yes	No	No		
0000.0014.0000.007.000.000.000.000.000	USB Input Device	HID (Human Interface Device)	No	Yes	No	No		
Dell KB216 Wired Keyboard	USB Composite Device	Unknown	Yes	Yes	No	No		
Dell KB216 Wired Keyboard	USB Input Device	HID (Human Interface Device)	Yes	Yes	No	No		
Dell KB216 Wired Keyboard	USB Input Device	HID (Human Interface Device)	Yes	Yes	No	No		
Dell MS116 USB Optical Mouse	USB Input Device	HID (Human Interface Device)	Yes	Yes	No	No		
Port_#0001.Hub_#0001	HP USB Flash Drive USB Device	Mass Storage	No	Yes	No	No		070D2948AEFF8325
Port_#0001.Hub_#0001	HP USB Flash Drive USB Device	Mass Storage	No	Yes	No	No		07002B78AC84CD78
Port_#0001.Hub_#0001	HP v236w USB Device	Mass Storage	No	Yes	No	No		5619279D3F4B07A1

bible.usccb.org

ENG
IN
05:58 PM
15-10-2026

Result

The USB forensic examination using USBDeview was successfully executed, and historical USB connection artifacts were analyzed.

EXPERIMENT 05

View Last Activity of Your PC

Aim

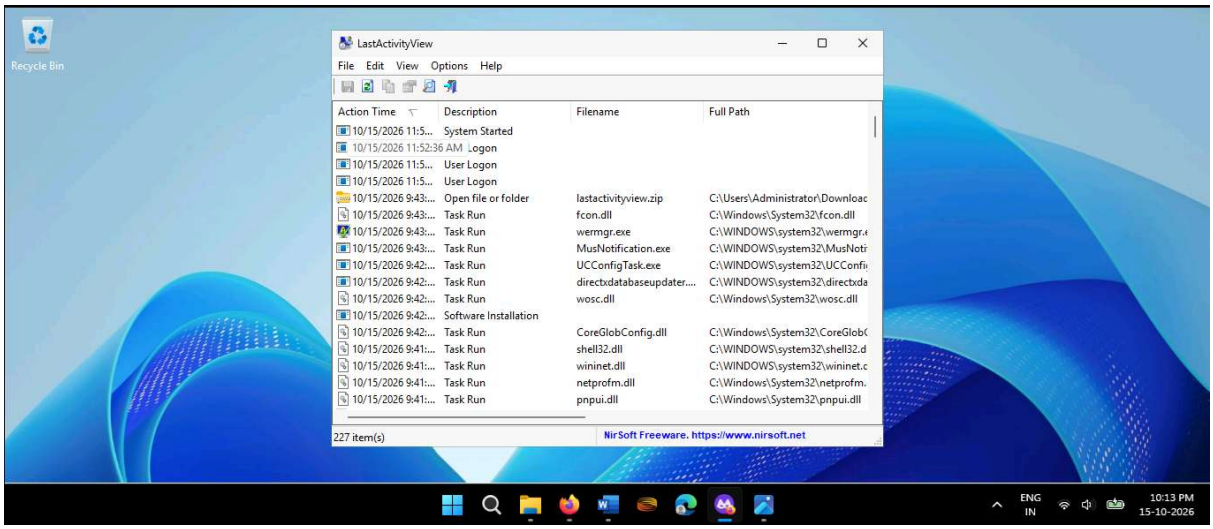
To collect and analyze system logs to view the detailed history of user actions and computer events on a running Windows OS.

Software / Tools Required

- LastActivityView Utility
- Windows Host PC

Procedure

1. Launched **LastActivityView** on the active Windows host.
2. Allowed the utility to gather event records from system sources (Registry, log files, system folders).
3. Filtered and reviewed the chronological log entries showing user activities, app executions, and system events.



Result

System activity forensic examination using LastActivityView was successfully performed and historical logs were analyzed.

EXPERIMENT 06

Extraction of Browser Artifacts

Aim

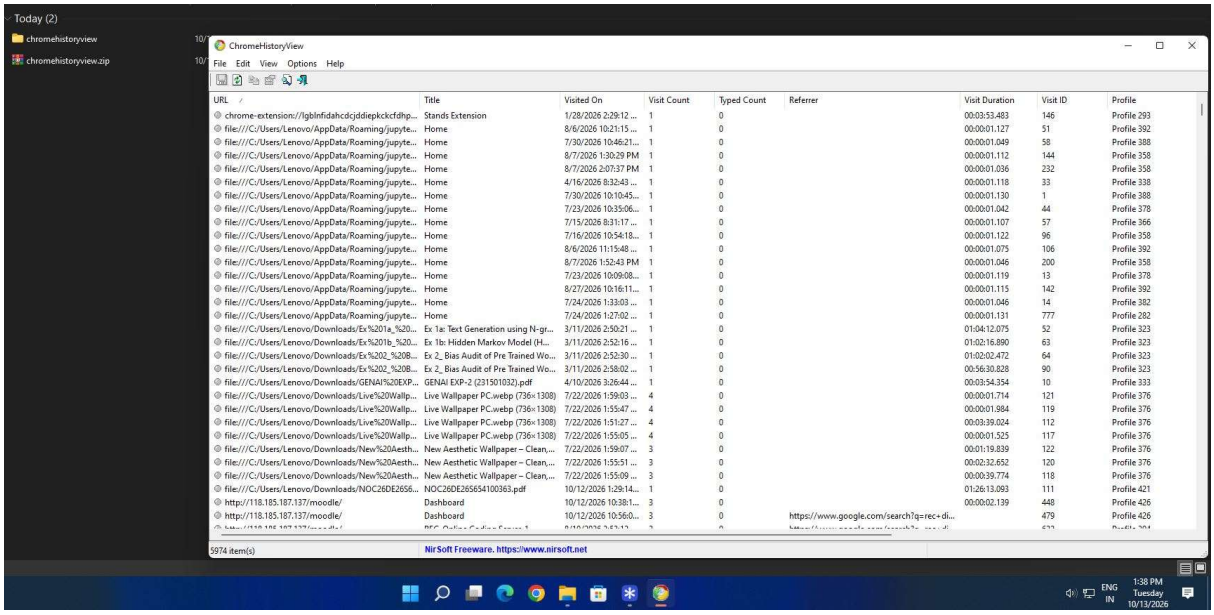
To extract and analyze web browser history, cache, and artifact data using specialized forensic utilities.

Software / Tools Required

- ChromeHistoryView & ChromeCacheView
- IEHistoryView & IECacheView
- Windows Host PC with Google Chrome and Internet Explorer installed

Procedure

1. **Chrome Extraction:** Launched **ChromeHistoryView** to view browsing history files and **ChromeCacheView** to inspect cache files.
2. **IE Extraction:** Executed **IEHistoryView** to read visited URL records and **IECacheView** to inspect cached file details.
3. **Data Analysis:** Evaluated browsing attributes (URLs, access timestamps, file paths, visit counts) and exported entries to HTML/text reports.



Result

Browser forensic tools were executed successfully, and browser artifacts were accurately extracted and analyzed.

EXPERIMENT 07

Comparison of Two Files for Forensics Investigation by Compare IT Software

Aim

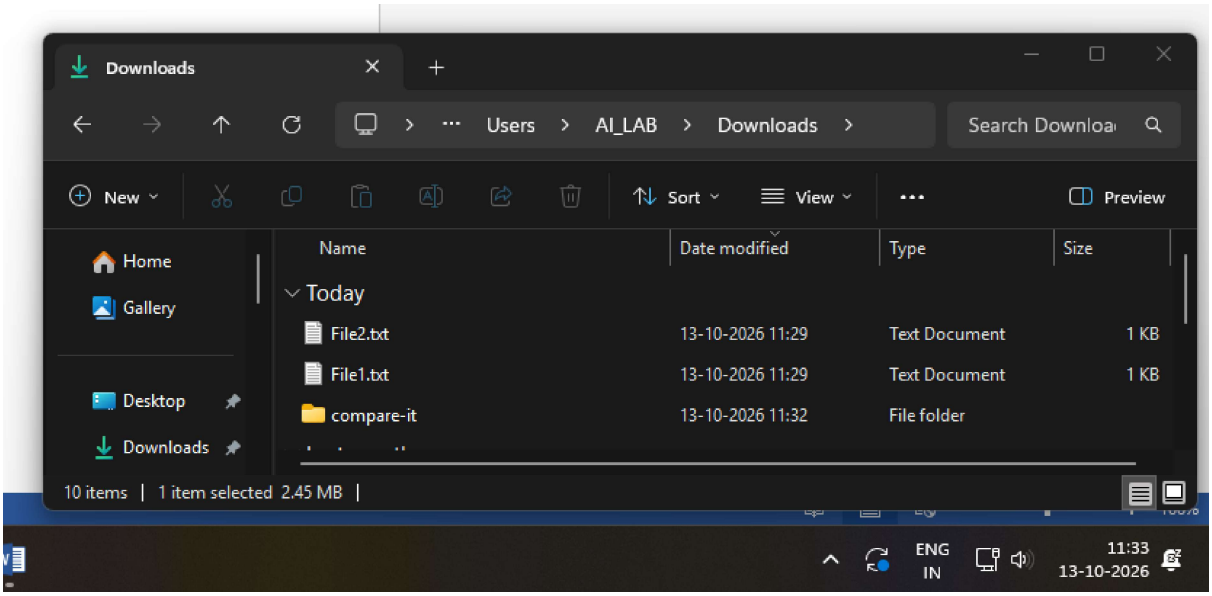
To compare two files side-by-side using **Compare IT** software to detect structural and content differences during a digital forensic investigation.

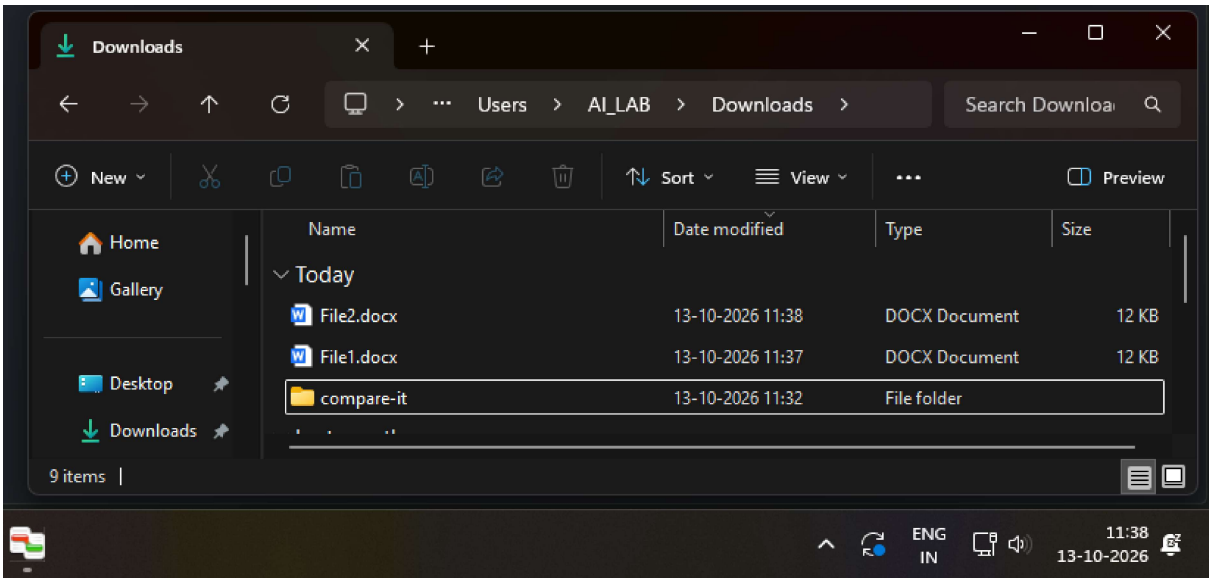
Software / Tools Required

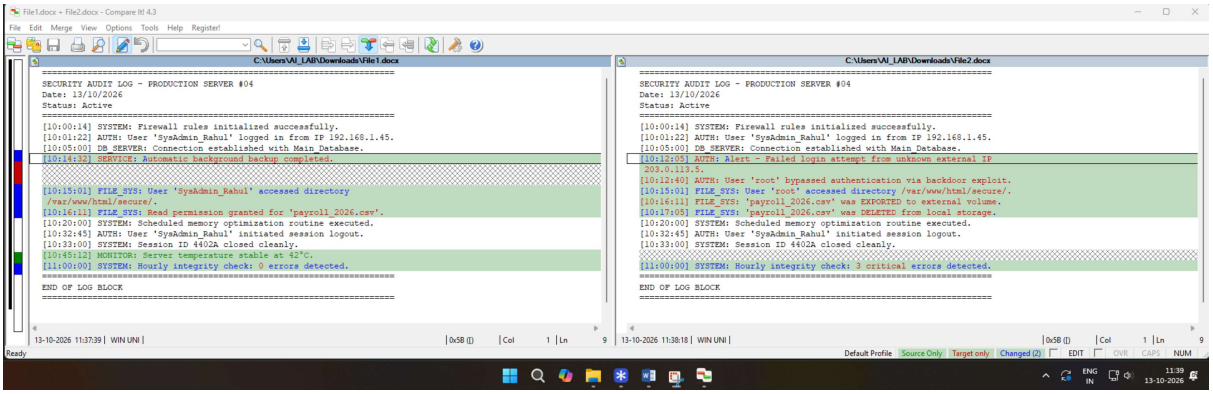
- Compare IT Software
- Target files for comparison

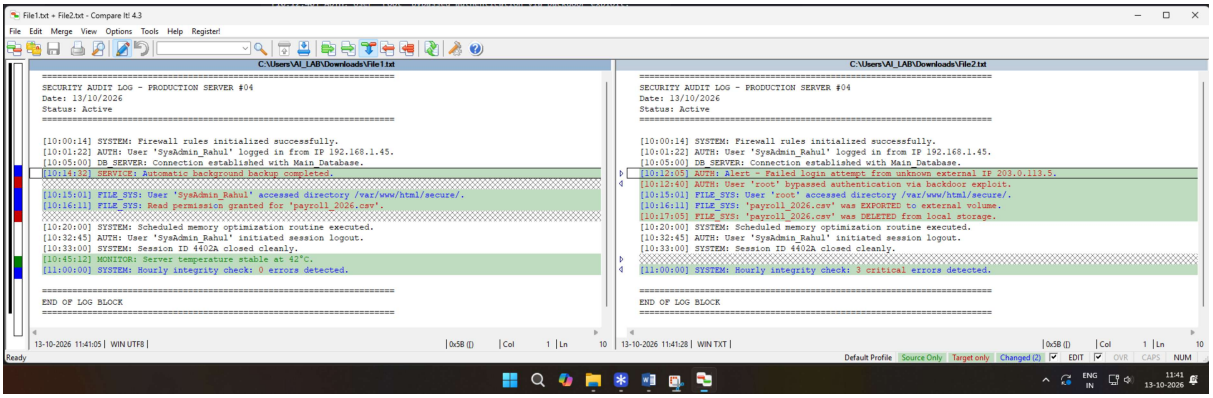
Procedure

1. Installed and opened the **Compare IT** application.
2. Loaded the first file into the left pane and the second file into the right pane.
3. Reviewed the side-by-side display with color-coded highlighted differences.
4. Saved/printed the color-coded difference report summarizing the findings.









Result

File comparison using Compare IT software was successfully conducted, and differences between target files were documented.

EXPERIMENT 08

Collecting Email Evidence in Victim PC

Aim

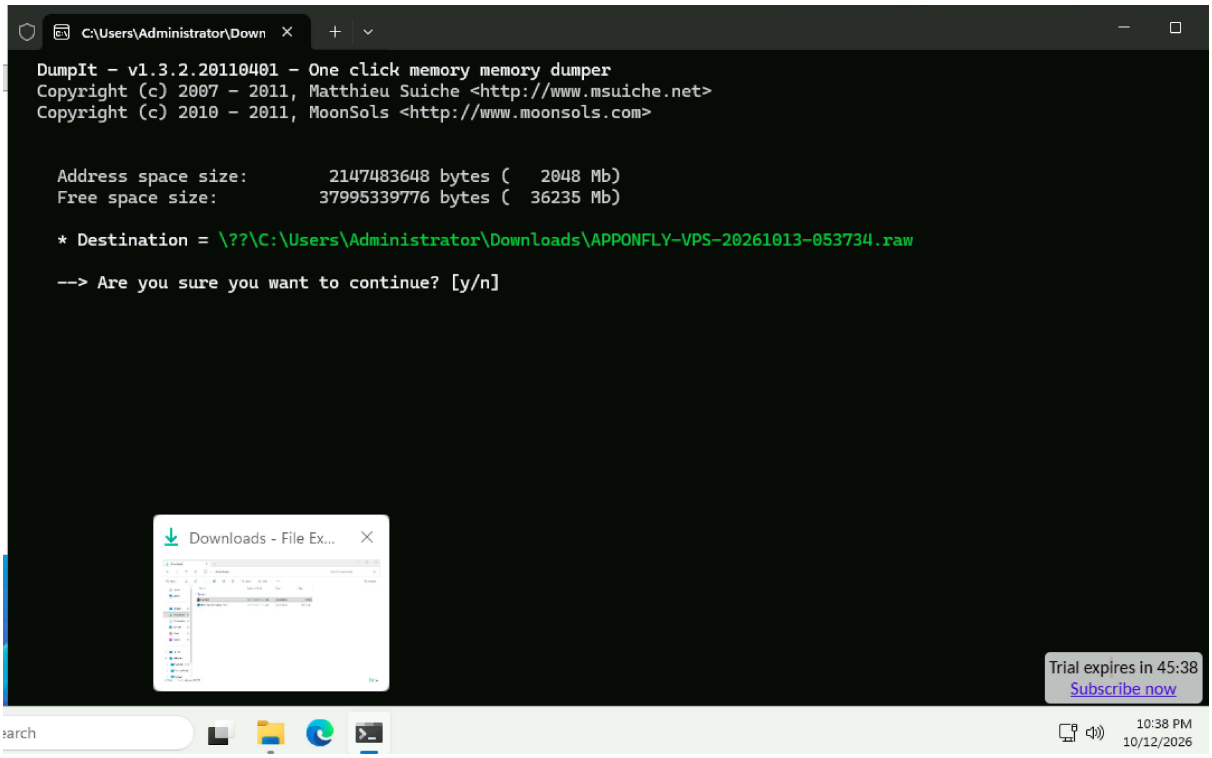
To collect and extract email evidence and volatile data from a victim PC by capturing and analyzing physical memory (RAM).

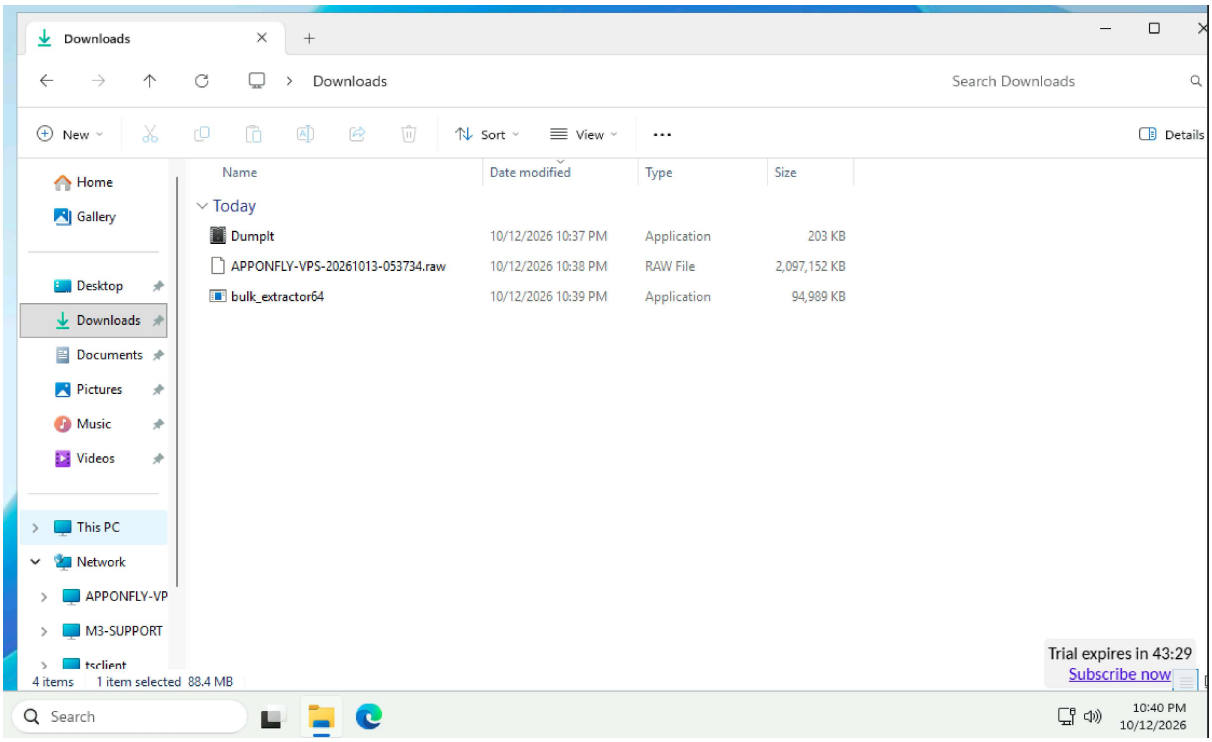
Software / Tools Required

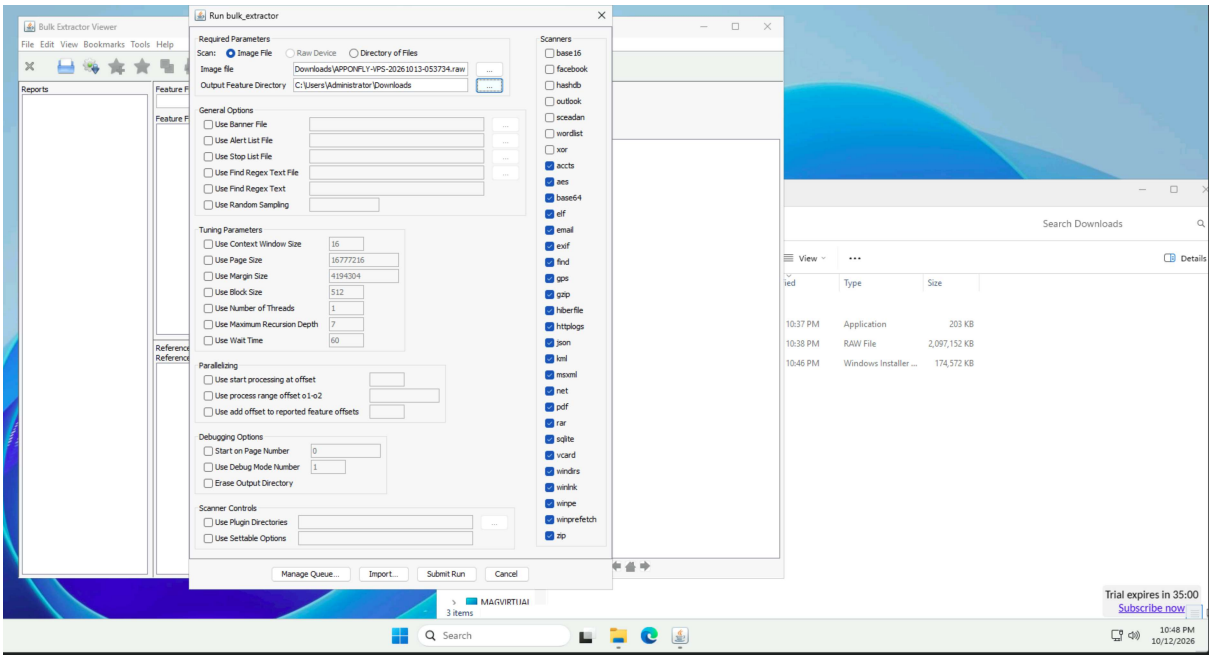
- DumpIt Utility
- Bulk Extractor (or RAM analyzer)
- Target Windows Host PC

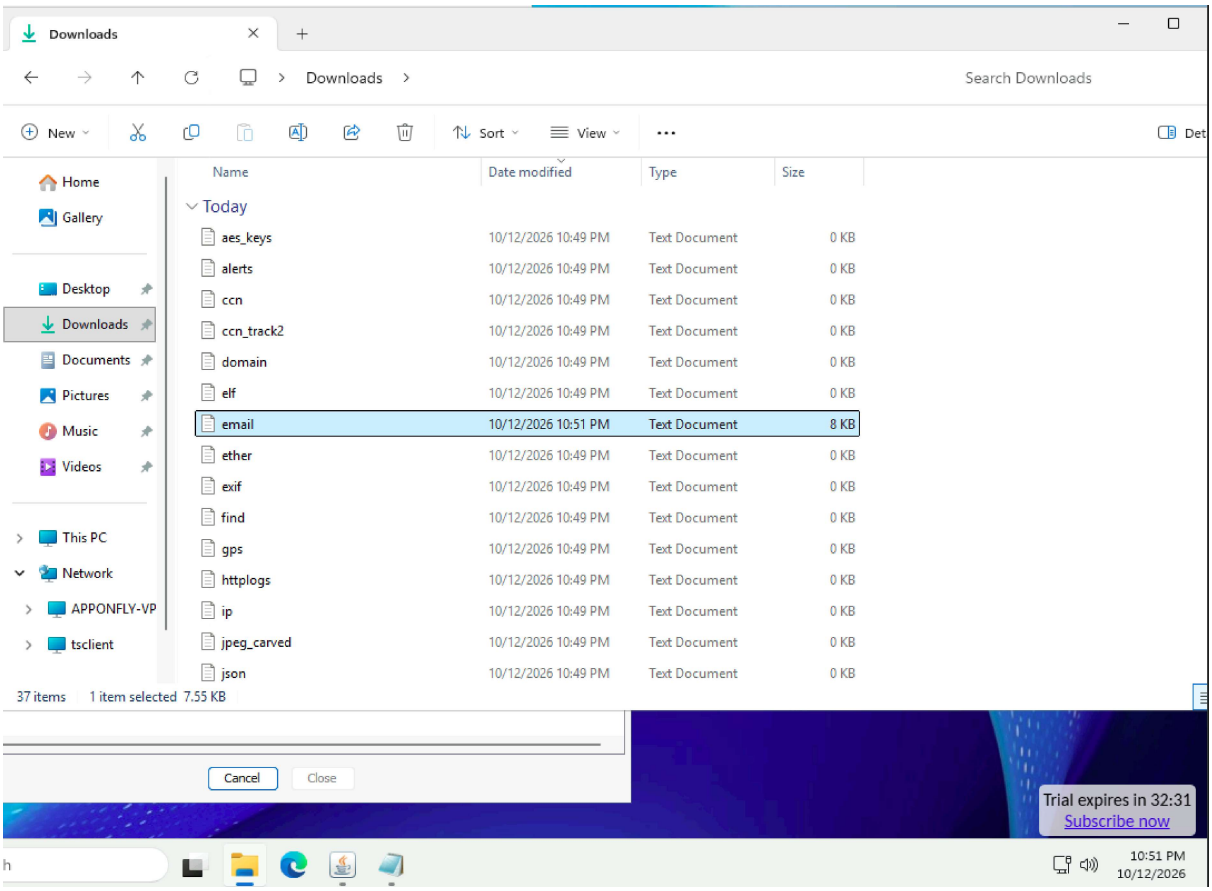
Procedure

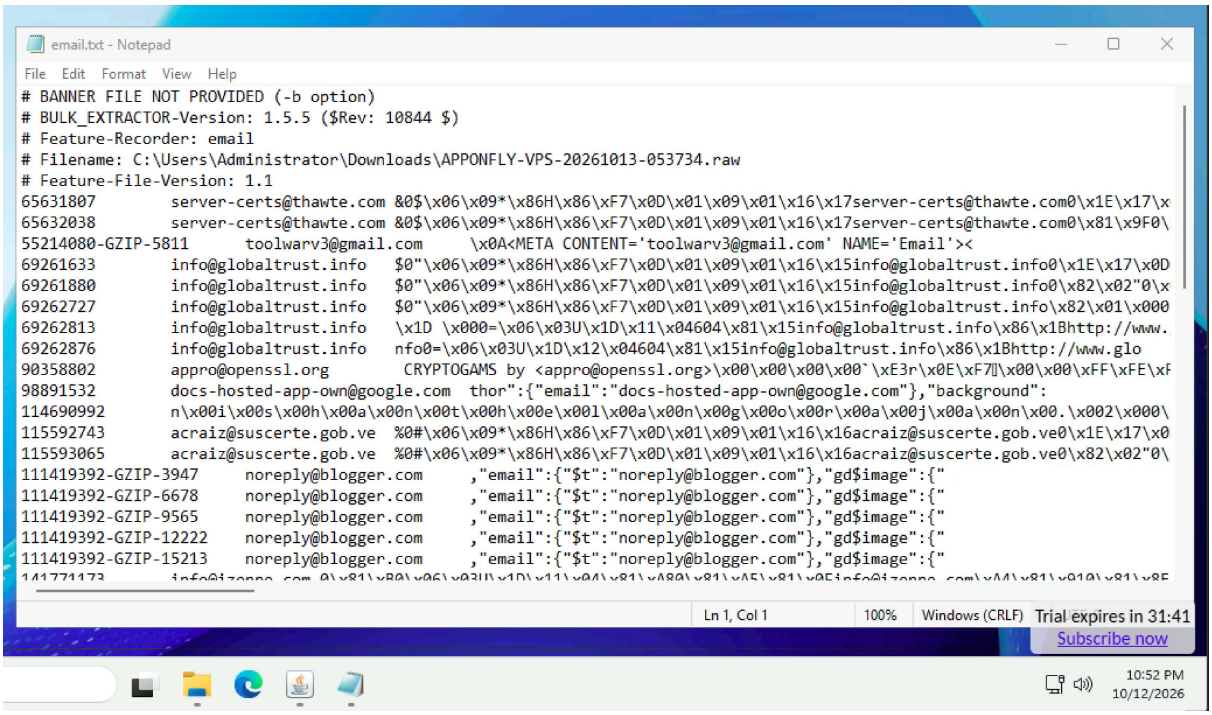
1. Executed **DumpIt.exe** directly from an external USB drive on the victim machine.
2. Confirmed the execution prompt to generate a full, raw physical memory dump (.raw file).
3. Used **Bulk Extractor** to scan the generated RAM image for email addresses, headers, and message fragments.




Downloads
Search Downloads
New
Sort
View
Details
Home
Gallery
Desktop
Downloads
Documents
Pictures
Music
Videos
This PC
Network
APPONFLY-VP
M3-SUPPORT
Name
Date modified
Type
Size
Today
Dumpit
10/12/2026 10:37 PM
Application
203 KB
APPONFLY-VPS-20261013-053734.raw
10/12/2026 10:38 PM
RAW File
2,097,152 KB
bulk_extractor64
10/12/2026 10:39 PM
Application
94,989 KB
4 items
1 item selected 88.4 MB
Trial expires in 43:29
Subscribe now
Search
10:40 PM
10/12/2026


Bulk Extractor Viewer
Run bulk_extractor
Required Parameters
Scanners
General Options
Tuning Parameters
Parallelizing
Debugging Options
Scanner Controls
Manage Queue...
Import...
Submit Run
Cancel
Trial expires in 35:00
Subscribe now
Search
10:48 PM
10/12/2026





Result

Physical RAM from the victim PC was successfully dumped using DumpIt, and email evidence was accurately extracted.

EXPERIMENT 09

Hide and Extract Text File Behind an Image/Audio File Using Command Prompt

Aim

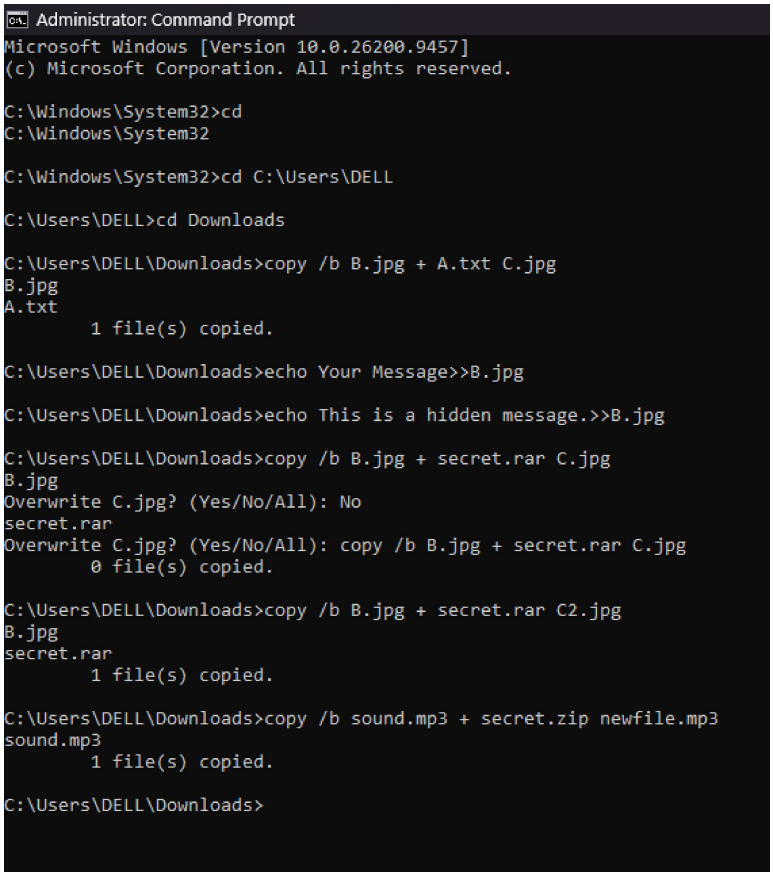
To study the steps for hiding and extracting a text file inside an image file using the Windows Command Prompt (Steganography).

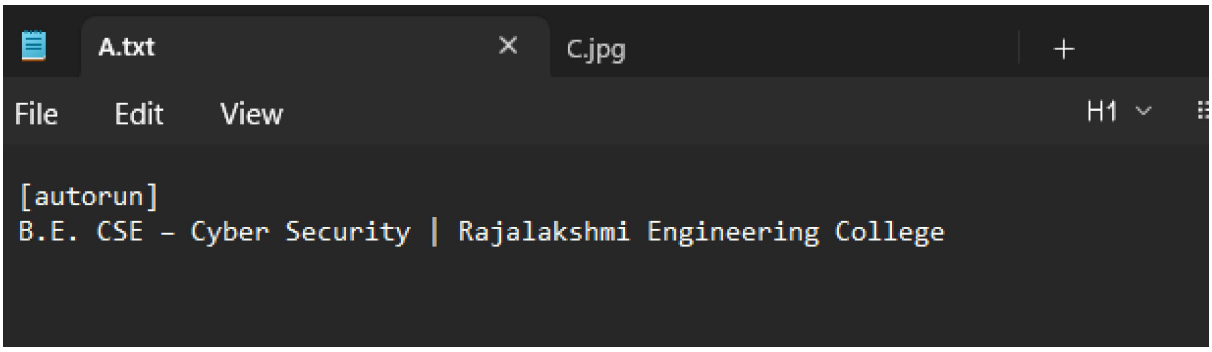
Software / Tools Required

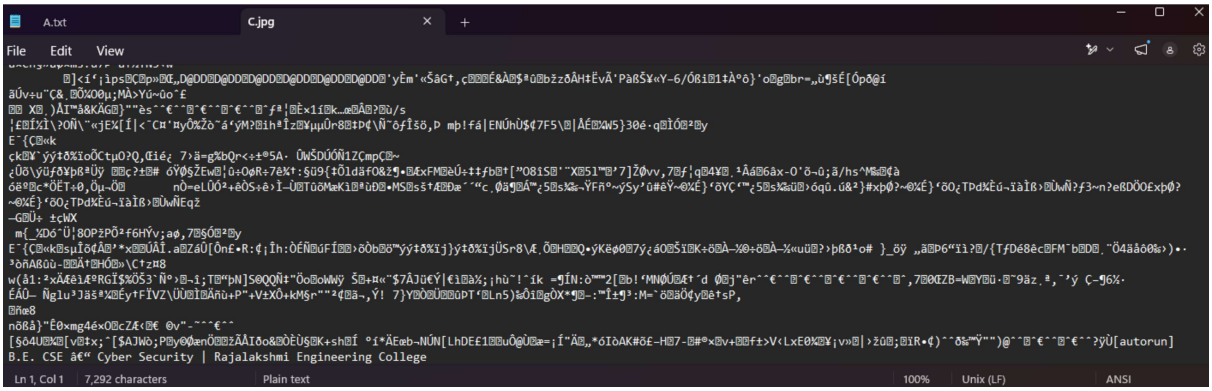
- Command Prompt (cmd.exe)
- Source Image file (B.jpg)
- Text file containing secret data (A.txt)
- WinRAR / Archiving tool

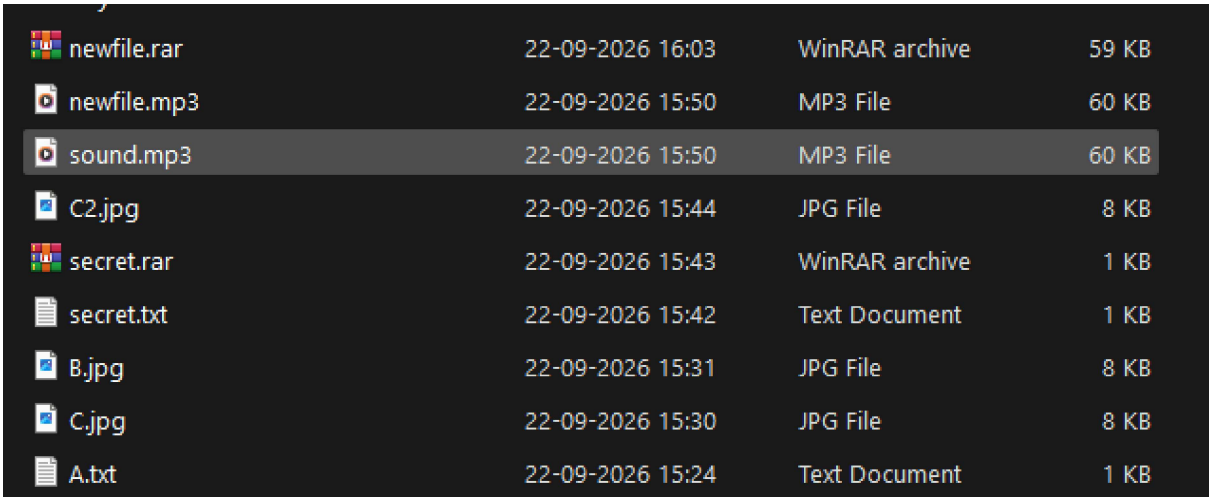
Procedure

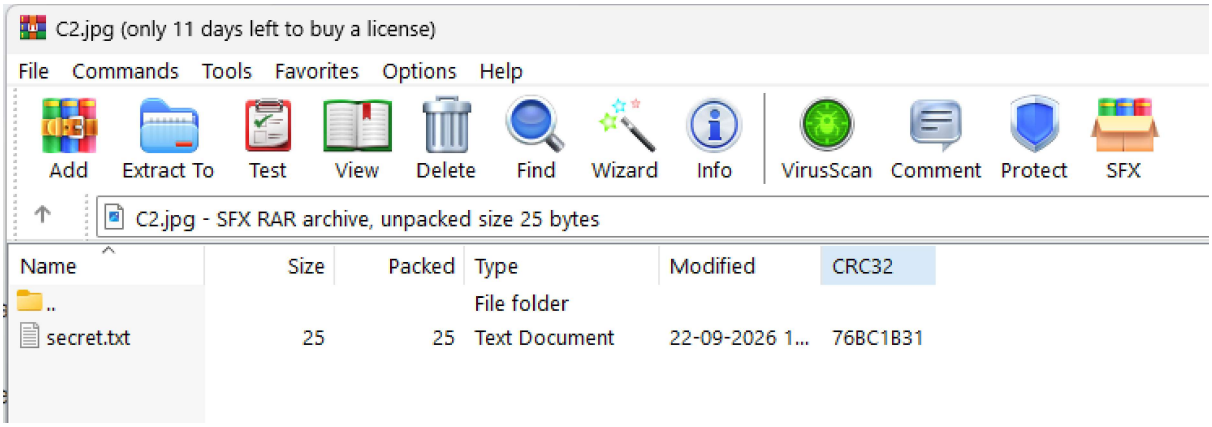
1. Placed cover image (B.jpg) and text file (A.txt) in the same folder directory.
2. Opened Command Prompt, navigated to the folder, and executed the binary copy command:copy /b B.jpg + A.txt C.jpg
3. Opened the output image (C.jpg) using WinRAR or command-line utilities to view and extract A.txt.

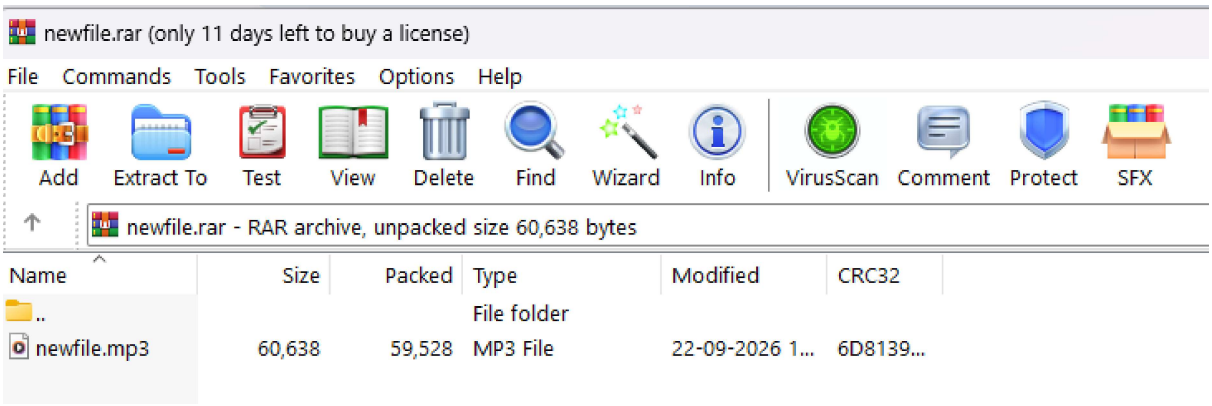



A.txt
C.jpg
File
Edit
View
H1
[autorun]
B.E. CSE – Cyber Security | Rajalakshmi Engineering College


A.txt
C.jpg
File
Edit
View
[autorun]
B.E. CSE – Cyber Security | Rajalakshmi Engineering College
Ln 1, Col 1
7,292 characters
Plain text
100%
Unix (LF)
ANSI


newfile.rar 22-09-2026 16:03 WinRAR archive 59 KB
newfile.mp3 22-09-2026 15:50 MP3 File 60 KB
sound.mp3 22-09-2026 15:50 MP3 File 60 KB
C2.jpg 22-09-2026 15:44 JPG File 8 KB
secret.rar 22-09-2026 15:43 WinRAR archive 1 KB
secret.txt 22-09-2026 15:42 Text Document 1 KB
B.jpg 22-09-2026 15:31 JPG File 8 KB
C.jpg 22-09-2026 15:30 JPG File 8 KB
A.txt 22-09-2026 15:24 Text Document 1 KB


C2.jpg (only 11 days left to buy a license)
File
Commands
Tools
Favorites
Options
Help
Add
Extract To
Test
View
Delete
Find
Wizard
Info
VirusScan
Comment
Protect
SFX
C2.jpg - SFX RAR archive, unpacked size 25 bytes
Name
Size
Packed
Type
Modified
CRC32
..
File folder
secret.txt
25
25
Text Document
22-09-2026 1...
76BC1B31



Result

The text file was successfully hidden behind the image file using Command Prompt, and the hidden data was retrieved.

EXPERIMENT 10

Extract Exchangeable Image File Format (EXIF) Data from Image Files using Exifreader Software

Aim

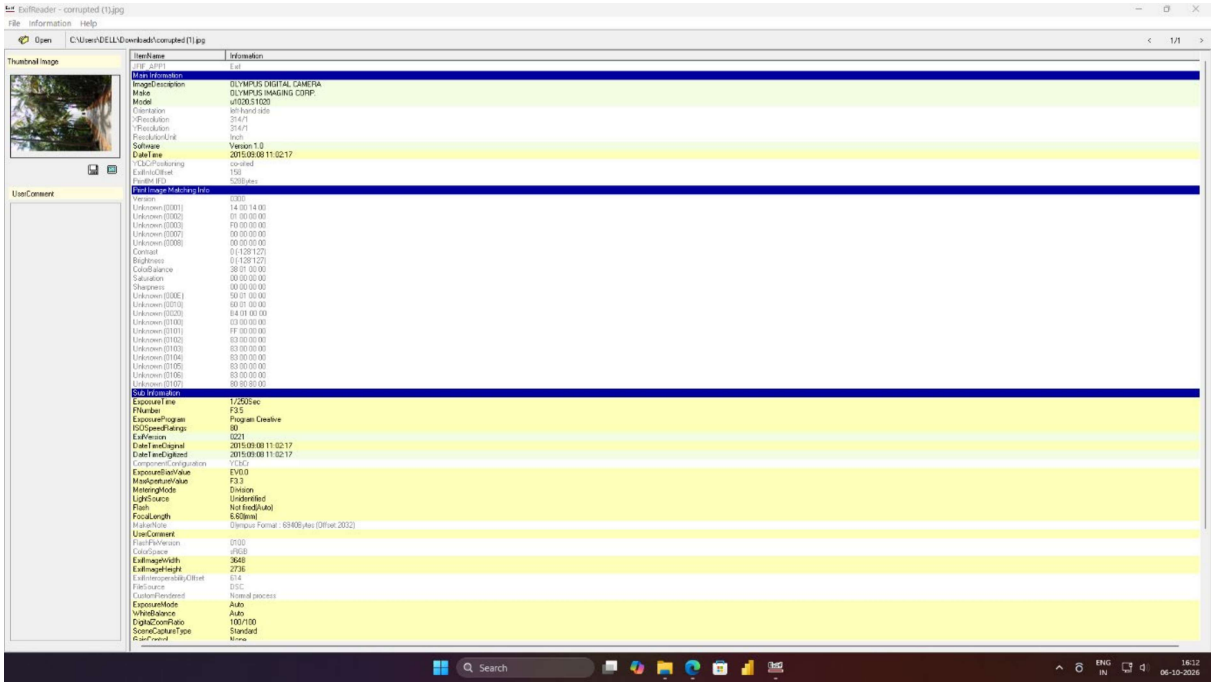
To extract and analyze metadata/EXIF data embedded within digital image files using **Exifreader** software for forensic investigation.

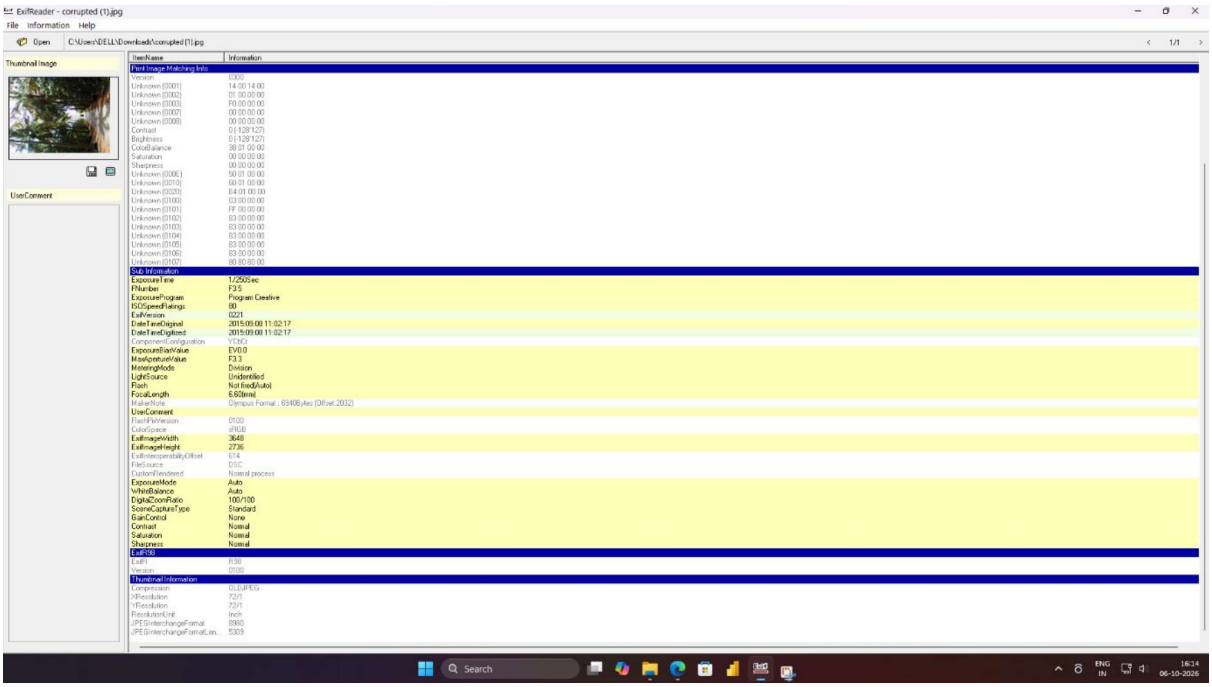
Software / Tools Required

- Exifreader Software
- Target Digital Image File (JPEG/TIFF image from a camera or smartphone)

Procedure

1. Opened the **Exifreader** application on the investigation host.
2. Imported the target digital image file into the software interface.
3. Allowed the tool to automatically parse embedded EXIF header tags.
4. Reviewed camera settings, device details, timestamps, and geolocation information.





Result

EXIF data was successfully extracted from the target image file using Exifreader, providing valuable metadata for linking evidence to a device and timeline.